Unifying Fairness Privacy Robustness and Explainability in Federated Graph Neural Networks for High Stakes Risk Detection

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Abstract

Federated Learning (FL) enables institutions to train shared models over sensitive, decentralised graph data—transactions, credit records, social ties—without exchanging raw data. Yet accuracy alone does not make such systems deployable: recent regulatory and scientific frameworks (the EU AI Act, the NIST AI Risk Management Framework) demand that automated decision systems be simultaneously *fair*, *private*, *robust*, and *transparent*. Existing work addresses these properties in isolation, and no prior method combines fairness, differential privacy, and Byzantine robustness for federated graph neural networks (GNNs). We present **TrustFedGNN**, a unified trustworthy federated graph-learning framework built from three components: (i) *Fairness-Sensitive Edge Reweighting* (FSER), which suppresses the biased cross-group message passing responsible for structural discrimination; (ii) *Fairness-Targeted Gradient Decomposition* (FTGD), which applies differential privacy to the *released fairness statistic* by privatising only the low-dimensional demographic-parity means—so strong DP for the fairness signal costs $\boldsymbol{O(1)}$ noise where full-gradient DP-SGD must inject $\boldsymbol{O(\sqrt{|\theta|})}$ and destroys utility. We emphasise at the outset that this $\boldsymbol{(\epsilon, \delta)}$ guarantee covers the *released fairness statistic only*, and *not* the full transmitted model update, which retains sensitive-attribute dependence through FSER and the fairness-gradient masks; the comparison against full-gradient DP-SGD is therefore not like-for-like, and we make the exact scope precise in Section 4.2. The framework further contributes (iii) *Bi-objective Frank–Wolfe Aggregation* (BFWA) and its Byzantine-robust variant, which impose an operator-chosen constraint on the aggregated client-disparity statistic while screening malicious clients, including a new fairness-poisoning threat. We further instrument the framework with predictive uncertainty, GNN attention audits, a composite trust score, sustainability accounting, and an EU AI Act / NIST RMF compliance mapping. Across five real datasets spanning credit, recidivism, social, and—uniquely— large-scale crypto (Elliptic Bitcoin, 203,769 nodes) domains, TrustFedGNN does not dominate every individual metric, but it is the only method among fifteen baselines to combine competitive fairness and utility with a statistic-level sensitive-attribute DP guarantee and Byzantine robustness in a single federated GNN. All tables and figures are regenerated from logged runs and the code is released, so every reported number is traceable to a configuration. This work supports SDG 9 by advancing trustworthy collaborative AI.

Keywords: Federated learning, Graph neural networks, Algorithmic fairness, Differential privacy, Byzantine robustness, Explainable AI, Trust metrics, Regulatory compliance

1 Introduction

Consider a consortium of banks that together want to catch money laundering. Fraud is fundamentally *relational*: illicit funds move through chains of accounts, shell companies, and intermediaries, and it is precisely the *structure* of these money flows—not any single transaction in isolation—that betrays criminal intent. Graph Neural Networks (GNNs) have therefore become the tool of choice for financial crime detection, credit scoring, and other high-stakes risk assessments, because their message-passing mechanism learns directly from the web of relationships between users, accounts, and transactions [1–3]. Yet the very banks that would benefit most from a shared model are legally forbidden from pooling their customers' data: privacy regulations such as the GDPR, and plain commercial confidentiality, keep each institution's transaction graph locked inside its own walls. Federated Learning (FL) offers a way out. By exchanging model updates instead of raw data, a group of institutions can train a single, more powerful detector without any of them surrendering its records [4].

This picture, however, is dangerously incomplete. A model that is merely *accurate* is no longer fit to be deployed in decisions that shape people's access to credit, liberty, or financial services. Three converging forces—the European Commission's Ethics Guidelines for Trustworthy AI, the EU AI Act, and the U.S. NIST AI Risk Management Framework—now demand that automated decision systems be—depending on their risk classification, the provider's role, and the jurisdiction—*fair*, *privacy-preserving*, *robust*, *transparent*, and *accountable*. For systems that fall under the EU AI Act's high-risk category, several of these move from good practice toward regulatory obligation (subject to the Act's scope and conformity-assessment provisions, which we do not attempt to adjudicate here). Meeting them is hard even for a centralised model. In the federated, graph-structured setting they collide, and each one becomes markedly harder than in isolation.

Why the federated graph setting is uniquely hard.

The difficulty is not that we must bolt four independent features onto a classifier; it is that the natural solution to each requirement actively sabotages the others. *Fairness* is confounded by the graph itself: homophilous message passing means that a member of a demographic group who happens to be connected to risky neighbours will be flagged as risky too, so the model learns to launder a protected attribute through graph structure—and the central server, which never sees any data, cannot even measure this bias, let alone correct it. *Privacy* appears to have a textbook answer, differentially private SGD, but that answer fails badly here: adding calibrated noise to every coordinate of the gradient injects perturbation whose magnitude grows with the square root of the model dimension, and on the compact GNNs used for tabular financial graphs this noise simply drowns the signal, collapsing accuracy to little better than chance (we quantify this in Section 5.6). *Robustness* is threatened not only by the classical malfunctioning or malicious client, but—as we show—by a subtler adversary that the fairness machinery itself invites: a client can transmit a deliberately biased model while *reporting* exemplary fairness statistics, thereby hijacking any aggregator that rewards fairness. And *transparency* is undermined by the opacity of relational models, exactly when regulation requires that each adverse decision be explainable and auditable.

The gap.

The research literature has, understandably, attacked these problems one at a time. Fair-GNN methods deliver impressive demographic parity but assume all data sits in one place [5–7]. Fair federated learning has matured for tabular and image data but rarely touches graphs [8, 9]. Byzantine-robust aggregation hardens FL against poisoning but is oblivious to fairness [10, 11]. Differentially private FL protects data but, as noted, at a utility cost that is prohibitive for graph models. To the best of our knowledge, *no* existing method delivers fairness, differential privacy, and Byzantine robustness together for federated GNNs, and none has been validated on a large-scale cryptocurrency graph—arguably the most consequential modern fraud domain. Closing this gap is the goal of this paper.

Our approach.

We present **TrustFedGNN**, a single framework in which fairness, privacy, robustness, and transparency are designed to reinforce rather than undermine one another. Its architecture (Fig. 1) intervenes at the three natural control points of federated graph training—inside message passing, at the local gradient, and at server aggregation—and wraps the result in an instrumentation layer that measures, explains, and audits trustworthiness. The key technical insight, and the reason privacy becomes almost free, is a change of perspective on *where* the sensitive attribute actually influences learning: it does so only through a two-dimensional summary statistic, so we can protect that statistic directly instead of blindly noising the entire gradient.

Concretely, our contributions are:

1. **Fairness-Sensitive Edge Reweighting (FSER)**, a learnable attention correction that detects and suppresses the biased cross-group edges responsible for structural discrimination, mitigating unfairness at its source inside the GNN rather than patching it after the fact (Section 4.1).
2. **Fairness-Targeted Gradient Decomposition (FTGD)**. We observe that the sensitive attribute enters the *fairness objective* only through two demographic-parity means, and exploit this to release those bounded-sensitivity statistics under (ϵ, δ)-differential privacy—$O(1)$ noise instead of the $O(\sqrt{|\theta|})$ noise a full-gradient mechanism needs, precisely where DP-FedAvg collapses utility. We are explicit that this privatises the released fairness statistic, not the entire model update (which retains sensitive-attribute dependence through FSER and the fairness-gradient masks), and we prove the statistic-level guarantee with a self-contained Rényi-DP accountant (Section 4.2).
3. **Bi-objective Frank–Wolfe Aggregation (BFWA)** and a **Byzantine-robust variant**. The server solves a constrained optimisation that steers aggregation toward an operator-chosen target on the reported client-disparity statistic (a computable surrogate, not a certificate of global model fairness), while distance-based screening defends against poisoning— including the fairness-poisoning attack of [12], which we reproduce and defend against (Sections 4.3, 5.8).
4. **A trustworthiness instrumentation layer** that turns abstract principles into measured, auditable artifacts: predictive uncertainty and calibration, GNN attention audits with fairness attribution, a transparent composite *trust score* (with its sensitivity to reference constants stated openly), sustainability accounting, and an explicit EU AI Act / NIST RMF design-mapping with an auto-generated model card (Sections 4.5, 6).
5. **A reproducible evaluation** on five real datasets spanning credit, recidivism, social, and—uniquely—large-scale crypto (Elliptic Bitcoin, 203,769 transactions) domains, against fifteen baselines that include the most recent 2025–2026 fairness-aware federated and graph methods. Not every baseline is run on every dataset—cost and scale restrict the Credit and Elliptic columns, as Section 5.1 states explicitly—and TrustFedGNN does not win every metric, but it uniquely combines competitive fairness/utility with statistic-level sensitive-attribute DP and Byzantine robustness, and every number and figure in this paper is regenerated from logged runs by a single script.

Relation to a preliminary version.

A short preliminary version of this work, introducing the FSER, FTGD, and BFWA components under the name FedFairGNN, appeared as a conference paper [13] evaluated on three fraud-detection graphs (YelpChi, Amazon, Elliptic) with a single K=3-client configuration and no robustness, uncertainty, explainability, or regulatory-compliance analysis. This article substantially extends that preliminary study: it (i) evaluates on five standard fairness benchmarks (German, Credit, Bail, Pokec, Elliptic) plus a 2.4M-node scalability study (ogbn-products), only one of which overlaps with the earlier evaluation; (ii) compares against fifteen baselines, including five 2025–2026 methods reimplemented for this comparison, versus the small baseline set of the preliminary version; (iii) introduces the Byzantine-robust `robust_bfwa` variant and a new fairness-poisoning attack, entirely absent from the earlier work; (iv) adds an adversarial evaluation of the privacy mechanism (an attribute-inference attack, Section 5.6) and multi-seed statistical significance testing, neither of which the preliminary version performed; and (v) adds the full trustworthiness instrumentation layer—predictive uncertainty and calibration, attention-based audits, a composite trust score, sustainability accounting, and the EU AI Act / NIST AI RMF design mapping (Section 6)—which did not appear in the preliminary version at all. We also renamed the framework from FedFairGNN to TrustFedGNN to reflect this broadened scope.

By lowering the barrier to trustworthy, infrastructure-grade collaborative AI, this work contributes to UN Sustainable Development Goal 9 (industry, innovation, and resilient infrastructure). The remainder of the paper is organised as follows. Section 2 situates our work within the fairness, privacy, and robustness literatures; Section 3 formalises the federated graph setting and the trustworthiness objectives; Section 4 develops the TrustFedGNN framework component by component; Section 5 reports the empirical study; Section 6 analyses trustworthiness holistically and maps the system to regulation; and Section 7 discusses implications, limitations, and broader impact before we conclude.

2 Related Work

Fairness in graph neural networks.

GNNs inherit and amplify bias through homophilous message passing. Remedies include adversarial debiasing (FairGNN [5]), counterfactual and stability regularisation (NIFTY [14]), pre-processing that minimises inter-group distributional distance (EDITS [15]), sensitive-information neutralisation via heterogeneous neighbours (FairSIN [6]), re-balancing through counterfactual mixup (FairGB [16]), invariant learning across inferred sensitive partitions (FairINV [17]), and disentangled debiasing (DAB-GNN [7]). All assume *centralised* data. Our FSER is closest in spirit to the edge-reweighting variants of FairSIN and to distribution-aware reweighting [18], but operates inside a federated, private, and robust pipeline.

Fairness in federated learning.

Two notions coexist. *Client (performance) fairness* equalises accuracy across clients: q-FedAvg [9] up-weights high-loss clients. *Group fairness* equalises outcomes across demographic groups: FairFed [8] adjusts aggregation weights by each client's local fairness gap; FedFB ports FairBatch to FL; PUFFLE [19] blends a demographic-parity loss with DP-SGD under a target-disparity controller; FedFACT [20] gives provable global-and-local group fairness via a Lagrangian saddle point over a shared global dual variable and per-client local duals; Bendoukha et al. [21] (PoPETs'25) make FairFed's weighting homomorphically computable, replacing its $\exp(-\beta|\cdot|)$ term with an FHE-friendly degree-2 polynomial and aggregating under threshold CKKS; and

EquFL [22] (2026) adds a server-side fairness-gradient correction on top of an arbitrary aggregation rule, provably compatible with Byzantine-robust ones. These target tabular or image data. For *graphs*, F^2GNN [23] combines a fairness-weighted aggregation with structure-aware group-balance weights. The most recent efforts sit closest to our setting: FairGFL [24] adds DP to federated fair GNNs but targets *cross-client* (performance) fairness arising from imbalanced overlapping subgraphs rather than demographic group fairness; FedGraph-Fair [25] applies distributionally-robust optimisation for personalised group-fair federated graph learning; FaVGNN [26] (2026) achieves group fairness in *vertical* FL via completion-driven adversarial fusion; and FDP-Fair [27] (2026) combines federated DP training with demographic-parity post-processing. A preliminary version of this work by two of the present authors [13] introduced the FSER, FTGD, and BFWA components under the name FedFairGNN on a smaller evaluation without a robustness, uncertainty, or compliance analysis; this article is a substantial extension, detailed in Section 1. We compare against FairFed, q-FedAvg, F^2GNN, FairSIN (run federated), the strongest 2026 competitors (horizontal adaptations of FaVGNN and FDP-Fair), and—reimplementing each paper's core statistical mechanism to the extent it transfers to a single-graph, star-topology federation (Appendix B)—FairGFL, FedGraph-Fair, PUFFLE, FedFACT, and the PoPETs'25 FairFed variant, none of which had a public head-to-head against a fairness-aware federated *GNN* prior to this comparison.

Differential privacy in FL.

DP-FedAvg [28] clips per-client updates and adds Gaussian noise; privacy is tracked with the moments accountant / Rényi DP [29, 30]. On the compact GNNs used for tabular graphs, isotropic gradient noise at meaningful ϵ overwhelms the signal (Section 5.6). FedFDP [31] adapts clipping to fairness but still privatises the whole gradient, and CE-FedGNN [32] (2026) instead targets communication efficiency under a metric-DP guarantee for federated GNN training—neither touches the fairness statistic specifically. FTGD instead privatises only the *sufficient statistics* of the fairness term, exploiting the fact that the sensitive attribute never enters the task loss—a targeted mechanism whose (ϵ, δ) guarantee follows from the Gaussian mechanism and post-processing immunity. A further 2026 result is cautionary: locally private GNNs remain vulnerable to structure-inference adversaries even under a formal ϵ budget [33], which is why we couple FTGD's privacy guarantee with the robust aggregator (BFWA/robust-BFWA) rather than treating DP alone as a sufficient defence.

Byzantine-robust aggregation.

Krum and Multi-Krum [10], coordinate-wise median and trimmed mean [11], Bulyan [34], and trust-bootstrapping schemes such as FLTrust [35] defend against malicious updates under attacks including sign-flipping, model-replacement [36], ALIE [37], and IPM [38]. None targets fairness. Recently, fairness-poisoning attacks emerged—EAB-FL [39] and the fairness-constrained optimisation attack [12]—that *increase* disparity while preserving accuracy, evading both robust and fairness-aware defences. We reproduce the fairness-constrained optimisation attack of [12] and defend with a robust fairness-constrained aggregator. Two further 2026 advances harden aggregation against stronger threat models without touching fairness: CQSA [40] clusters updates before a quantum-secure aggregation step, and a post-quantum lattice-based scheme [41] combines adaptive weighting with CRYSTALS-Kyber encryption for critical-infrastructure deployments; both are, like their predecessors, fairness-blind.

Explainability, uncertainty, and trust metrics.

GNN explainers attribute predictions to edges/features; we use the FSER attention itself as a native, inspectable audit signal (not a validated prediction-level explanation) and add gradient-based attribution of the *disparity*. Predictive uncertainty via MC-dropout [42] and calibration (ECE, Brier) quantify confidence reliability; a concurrent 2026 approach, FedWQ-CP [43], instead calibrates conformal-prediction sets in a single round under dual (statistical and system) heterogeneity—a complementary, distribution-free alternative to our MC-dropout-based calibration that we discuss as future work (Section 7). Composite trust scoring for FL is nascent; we contribute a transparent power-mean aggregation over utility, fairness, privacy, robustness, and calibration. On the compliance side, CryptoFair-FL [44] (2026) uses homomorphic encryption to verify demographic parity and equalised odds without exposing sensitive attributes, explicitly mapping to EU AI Act Articles 10 and 13—independent confirmation that regulatory-grounded, cryptographically verifiable fairness auditing is an active 2026 research direction, though it does not integrate Byzantine robustness or an explainability layer as we do. Finally, we align the system to the EU AI Act and NIST AI RMF, operationalising accountability as an auditable artifact rather than prose.

Positioning.

Table 1 places the most relevant methods on the five axes that define our problem. The pattern is stark: the fair-GNN methods are strong on graph fairness but centralised; the fair-FL methods federate but ignore graph structure; robust aggregation hardens FL but is fairness-blind; and even the newest federated fair-graph methods (F^2GNN, FairGFL, FaVGNN, FedGraph-Fair) each leave at least one axis uncovered—typically Byzantine robustness, which no fairness-aware federated method addresses. Among the methods surveyed here, TrustFedGNN is the only entry that combines all five axes at once—federated, graph-native, group-fair, differentially private (for the fairness statistic), and Byzantine-robust—and the only one we evaluate on a large-scale cryptocurrency graph (using, as noted, a temporal proxy rather than a demographic attribute). It also reports the fullest trustworthiness picture—uncertainty, attention audits, sustainability, and a regulatory design-mapping—within a single framework. This axis-coverage claim is about *breadth of integration*, not about being best on any single axis, where, as Section 5.3 shows, specialised baselines can win individual cells.

3 Preliminaries and Problem Setting

Cross-silo federated graph learning.

There are K institutions (clients). Client k holds a private graph $\mathcal{G}_k = (\mathcal{V}_k, \mathcal{E}_k, \mathbf{X}_k, \mathbf{y}_k, \mathbf{s}_k)$ with node features $\mathbf{X}_k \in \mathbb{R}^{n_k \times d}$, binary task labels $\mathbf{y}_k \in \{0,1\}^{n_k}$

Table 1: Where methods sit across the trust axes. ✓: supported; ∼: partial/adaptable; blank: not addressed. “Graph” = graph-native; “Group fair” = demographic group fairness (not merely client fairness); “Crypto” = validated on a large-scale cryptocurrency graph. ‡Our DP entry is *not* equivalent to the record-level guarantees of DP-FedAvg, PUFFLE, FDP-Fair or FaVGNN: TrustFedGNN privatises the *released fairness statistic* rather than the whole model update (Section 4.2), so this column marks the presence of a formal (ϵ, δ) guarantee, not guarantees of equal scope.

Method	Fed.	Graph	Group fair	DP	Byz.	Explain	Crypto
FairGNN [5]		✓	✓			∼	
NIFTY [14]		✓	✓				
FairSIN [6]		✓	✓				
DAB-GNN [7]		✓	✓				
FairFed [8]	✓		✓				
q-FedAvg [9]	✓		∼				
PUFFLE [19]	✓		✓	✓			
FedFACT [20]	✓		✓				
DP-FedAvg [28]	✓			✓			
Krum/Median [10, 11]	✓				✓		
F^2GNN [23]	✓	✓	✓				
FairGFL [24]	✓	✓	∼	✓			
FaVGNN [26]	✓	✓	✓	✓			
FDP-Fair [27]	✓		✓	✓			
FedGraph-Fair [25]	✓	✓	✓				
PoPETs’25 [21]	✓		✓	✓			
TrustFedGNN (ours)	✓	✓	✓	✓‡	✓	✓	✓

(e.g. fraud/default/recidivism), and a binary *sensitive attribute* $\mathbf{s}_k \in \{0, 1\}^{n_k}$ (e.g. gender, age, race). Raw graphs never leave the client. A server coordinates T communication rounds; each round clients perform E local epochs and transmit model parameters, which the server aggregates. We evaluate the global model on the pooled held-out test nodes of all clients. Statistical heterogeneity is modelled by a symmetric Dirichlet(α) split over labels, the standard FL non-IID model [45]; smaller α is more skewed.

Group fairness.

For predicted probabilities $\hat{y}$ and sensitive attribute s we report three group-fairness metrics (lower is fairer):

$$\mathrm{DPD} = \big|\, \mathbb{E}[\hat{y} \mid s{=}0] - \mathbb{E}[\hat{y} \mid s{=}1] \,\big| \qquad \text{(Demographic Parity Difference)}, \tag{1}$$

$$\mathrm{EOD} = \big|\, \mathrm{TPR}_{s=0} - \mathrm{TPR}_{s=1} \,\big| \qquad \text{(Equal Opportunity Difference)}, \tag{2}$$

$$\Delta_{\mathrm{EOdds}} = \max\big(|\Delta\mathrm{TPR}|,\, |\Delta\mathrm{FPR}|\big) \qquad \text{(Equalized Odds)}. \tag{3}$$

DPD is computed on soft scores (matching the differentiable training term); EOD and Equalized Odds use thresholded predictions.

Differential privacy.

A randomised mechanism $\mathcal{M}$ is (ϵ, δ)-differentially private if for all adjacent datasets D, D' and all outputs S, $\Pr[\mathcal{M}(D) \in S] \leq e^{\epsilon} \Pr[\mathcal{M}(D') \in S] + \delta$ [46]. We protect the *sensitive attribute*: adjacency is defined by flipping one node's s_i. The Gaussian mechanism adds $\mathcal{N}(0, \sigma^2)$ noise calibrated to the L_2 sensitivity of the released quantity; composition across rounds is tracked with Rényi DP [29]. For the Gaussian mechanism with noise multiplier $z = \sigma/\Delta$, the Rényi divergence of order α is $\epsilon_{\mathrm{RDP}}(\alpha) = \alpha/(2z^2)$; over T releases it composes as $T\alpha/(2z^2)$, and the tight conversion to (ϵ, δ)-DP is $\epsilon = \min_{\alpha>1} \left[\epsilon_{\mathrm{RDP}}(\alpha) + \log(1/\delta)/(\alpha - 1)\right]$.

Threat model for robustness.

Up to $f < K/2$ clients are Byzantine and may send arbitrary updates. We consider data poisoning (label flipping), classical model poisoning (Gaussian, sign-flip, scaling/model-replacement [36]), the strong omniscient attacks ALIE [37] and IPM [38], and a *fairness-poisoning* attack: a malicious client transmits a biased, scaled update while reporting fabricated fairness/utility statistics (DPD ≈ 0, high AUC) to capture a fairness-aware aggregator. The server is honest but curious and never accesses raw data.

Objective.

We seek a global model that jointly (i) maximises task utility (AUC-ROC), (ii) minimises group unfairness, (iii) satisfies a target (ϵ, δ)-DP for the sensitive attribute, and (iv) retains utility and fairness under Byzantine corruption—the trustworthiness trade-off that motivates Section 4.

4 The TrustFedGNN Framework

The design philosophy behind TrustFedGNN is that trustworthiness should be engineered where each threat originates, not retrofitted afterwards. Structural bias is born inside message passing, so we correct it there; the sensitive attribute leaks through the gradient, so we privatise exactly the part of the gradient that carries it; bias and poisoning are amplified at aggregation, so we constrain and screen the aggregation step. The framework therefore places three lightweight interventions at the three natural control points of federated graph training and wraps them in a measurement layer that renders the result auditable. Figure 1 gives the end-to-end picture: in each round the server broadcasts the global parameters θ_{t-1}; every client runs E local epochs of an FSER-augmented GNN and, through FTGD, releases its fairness statistic under differential privacy (reducing, though not eliminating, the update's sensitive-attribute footprint—Section 4.2); the server then screens the updates for Byzantine behaviour and aggregates the survivors with BFWA under an operator-chosen fairness target to produce θ_t. We now develop each component, motivating it before formalising it.

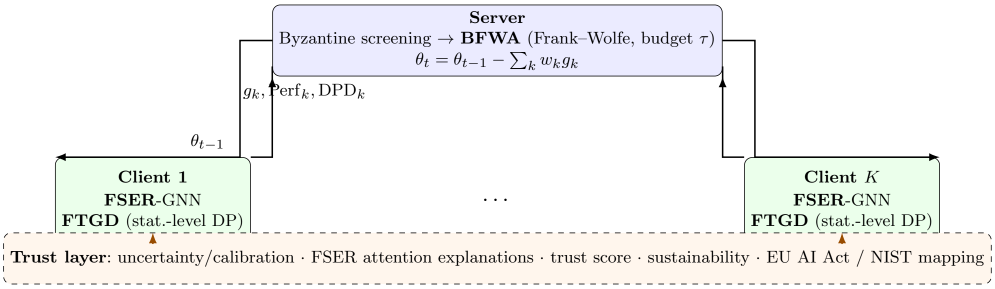

Fig. 1: TrustFedGNN architecture. Three interventions act at complementary control points—FSER inside client message passing, FTGD at the local gradient, and BFWA (with Byzantine screening) at server aggregation—while a trust layer instruments the whole system for audit. Only model weights are transmitted, so fairness and privacy add no communication overhead.

4.1 Fairness-Sensitive Edge Reweighting (FSER)

Intuition.

Message passing is the source of a GNN's power and, for fairness, of its peril. Each layer updates a node by mixing in its neighbours' representations, so a node inherits the statistical fate of the company it keeps. When the graph is homophilous along a protected attribute—members of one group disproportionately connected to one another, or to flagged neighbours—the network can encode group membership into embeddings, and the downstream classifier can exploit it without ever being told the sensitive attribute. Bias can propagate through both same-group and cross-group edges; FSER targets one specific, tractable pattern—a *cross-group* edge whose endpoints have become highly similar in representation space. Our hypothesis is that such edges disproportionately carry *demographic* rather than behavioural similarity, so down-weighting them reduces disparity at low utility cost. We do not claim to causally separate demographic from behavioural similarity—high cosine similarity can arise from genuinely shared behaviour, class, or neighbourhood— so FSER is best read as a *learnable heuristic* whose value is established empirically (Section 5.4), not as a certified causal debiaser. It is also, deliberately, only one of three interventions: FTGD and BFWA act on the pathways FSER leaves untouched.

Mechanism.

FSER adds a learnable correction to graph attention. For edge (i, j) with head embeddings $\mathbf{h}_i, \mathbf{h}_j$, the standard attention logit $e_{ij} = \mathrm{LeakyReLU}(\mathbf{a}^\top[\mathbf{h}_i \| \mathbf{h}_j])$ is corrected by a *fairness risk score*

$$\phi_{ij} = \mathbb{I}(s_i \neq s_j) \cdot \max\big(0, \cos(\mathbf{h}_i, \mathbf{h}_j)\big), \qquad \tilde{e}_{ij} = e_{ij} - \beta^{(l)}\phi_{ij}, \tag{4}$$

so that *cross-group* edges between similarly-embedded nodes are down-weighted after the per-node softmax. The suppression strength $\beta^{(l)} \geq 0$ is learned per layer (clamped to $[0, 5]$). Note that FSER evaluates $\mathbb{I}(s_i \neq s_j)$ and therefore requires the protected attribute of both edge endpoints at training *and* inference time: s is excluded from the node feature matrix $\mathbf{X}$ (so the classifier is not fed s as an input feature) but is used by the attention mechanism. This is a real deployment requirement—and a governance nuance we make explicit in Section 6—rather than a case of the model being blind to s. Because ϕ_{ij} and the resulting attention are retained, FSER also exposes an *inspectable audit signal*: Section 6 reports how much attention mass FSER removes from cross-group edges. We are careful not to call this a faithful explanation—attention weights are a diagnostic of the mechanism's behaviour, not a validated attribution of individual predictions (see the explainability discussion in Section 6).

4.2 Fairness-Targeted Gradient Decomposition (FTGD)

The DP-SGD dilemma.

Differentially private SGD is the standard way to keep a client's data from leaking through its updates, and in principle it is exactly what we need. In practice it is a poor fit here. DP-SGD clips the entire gradient to a norm bound C and adds isotropic Gaussian noise to every one of the $|\theta|$ coordinates; the total noise injected therefore grows like $\sigma\sqrt{|\theta|}$. For the large over-parameterised networks of vision and language this is tolerable, because the signal is spread across many redundant directions. The compact GNNs used on tabular financial graphs are the opposite regime: a few tens of thousands of parameters, every one of them carrying signal. At any privacy level strong enough to be meaningful, the noise required simply overwhelms the gradient, and accuracy collapses toward chance (Section 5.6 quantifies this). Blindly privatising the whole gradient is thus self-defeating. The question FTGD asks is: *do we actually need to?*

Objective.

The local objective is $\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda\,\mathcal{L}_{\text{fair}}$, with weighted BCE task loss and the differentiable soft demographic-parity penalty $\mathcal{L}_{\text{fair}} = |\mu_0 - \mu_1|$, where $\mu_g = \frac{1}{n_g}\sum_{i:s_i=g}\hat{y}_i$ is the group-g mean prediction.

Key observation.

The sensitive attribute s enters the fairness objective only through the two scalar group means (μ_0, μ_1): the task loss is a function of $(\mathbf{X}, \mathbf{y})$ alone, and s's contribution to $\mathcal{L}_{\text{fair}}$ is fully summarised by these two bounded-sensitivity statistics. FTGD therefore privatises *the released fairness statistic* with a Gaussian mechanism, so that strong DP for the fairness signal costs $O(1)$ noise rather than the $O(\sqrt{|\theta|})$ noise a full-gradient mechanism would inject. We are careful, below, to state precisely which released object this guarantee covers—and which sensitive-attribute pathways it does *not*.

Mechanism.

We treat the group counts (n_0, n_1) as public (they are needed to calibrate the noise scale, and each client's split enforces a public minimum group size $n_{\min}$). Under the adjacency that reassigns one node's group label with the counts held fixed, and using $\hat{y}_i \in [0,1]$, each mean changes by at most $1/n_g$, so (μ_0, μ_1) has L_2 sensitivity $\Delta =$

$\sqrt{1/n_0^2 + 1/n_1^2} \le \sqrt{2}/n_{\min}$. We release $\tilde{\mu}_g = \mu_g + \mathcal{N}(0, (z\Delta)^2)$ with noise multiplier z, form $\mathcal{L}_{\text{fair}} = |\tilde{\mu}_0 - \tilde{\mu}_1|$, and backpropagate. To reduce task/fairness gradient conflict we orthogonalise the task gradient against the fairness gradient (gradient surgery) before the step.

Proposition 1 (Privacy of the released fairness statistics) *Fix the public group counts* (n_0, n_1) *and the adjacency above. The per-step release of* $(\tilde{\mu}_0, \tilde{\mu}_1)$ *is the Gaussian mechanism at sensitivity* Δ *with noise* $z\Delta$*, hence* $(\alpha, \alpha/(2z^2))$*-RDP with respect to* s*. Over the* TE *releases the composed mechanism is* $(\alpha, TE\,\alpha/(2z^2))$*-RDP, and hence* $(\min_{\alpha>1}[TE\,\alpha/(2z^2) + \log(1/\delta)/(\alpha-1)], \delta)$*-DP, for the sequence of released statistics.*

Proof sketch Each release is the Gaussian mechanism at sensitivity Δ and noise $z\Delta$, giving noise multiplier z and RDP $\alpha/(2z^2)$ [29]. RDP composes additively across the TE releases and converts to (ϵ, δ)-DP by the standard bound. Any function of the released noisy statistics *alone* inherits the guarantee by post-processing. □

Scope and limitations of the guarantee.

We state plainly what Proposition 1 does and does not cover, because the distinction is easy to overclaim. The guarantee is on the *released group statistics* $(\tilde{\mu}_0, \tilde{\mu}_1)$ and on any quantity computed from them alone. It does *not* certify that the full transmitted update g_k is DP with respect to s: the fairness-gradient term $\nabla_\theta|\tilde{\mu}_0 - \tilde{\mu}_1| = \text{sign}(\tilde{\mu}_0 - \tilde{\mu}_1)\,(\nabla_\theta\mu_0 - \nabla_\theta\mu_1)$ still contains the raw group masks through $\nabla_\theta\mu_g$, and FSER's edge reweighting $\mathbb{I}(s_i \neq s_j)$ (Eq. 4) uses s directly in the forward pass. Adding noise to the *value* of the disparity randomises only its sign/magnitude, not the group-membership pattern that shapes the update direction. FTGD should therefore be read as *substantially reducing*, not eliminating, the sensitive-attribute footprint of the fairness signal: it removes the need to release an exact disparity and replaces the isotropic $O(\sqrt{|\theta|})$ noise of DP-SGD with $O(1)$ noise on the statistic that carries most of the signal. A fully sensitive-attribute-DP mechanism for the entire update would additionally have to privatise the fairness-gradient masks (e.g. a vector-valued Gaussian mechanism with proven gradient sensitivity) and FSER's use of s (or replace the hard indicator with a function of the noised statistics); we scope this stronger variant as future work (Section 7). Because z can be large while $z\Delta$ stays tiny (as $\Delta \sim 1/n_g$), the fairness statistic can be released at small ϵ with little utility loss—in contrast to full-gradient DP-SGD, whose isotropic noise scales with $\sqrt{|\theta|}$ (Section 5.6). We calibrate z once for the target ϵ over TE steps using the accountant above; consequently our privacy comparison against DP-FedAvg is *not* like-for-like (Section 5.6), since DP-FedAvg protects the whole record-level update while FTGD protects only the released fairness statistic.

4.3 Bi-objective Frank–Wolfe Aggregation (BFWA)

Why aggregation is a fairness lever.

FSER and FTGD act inside each client, but the client cannot see the global picture: a silo whose local data is demographically skewed will, in good faith, produce an update that pulls the global model toward its own bias, and plain FedAvg—which weights clients only by sample count—will faithfully propagate that bias to everyone. The server, by contrast, sees every client's reported fairness and does control the weights. This makes aggregation a natural *lever* on global fairness, and it lets a human operator set an explicit, auditable target on how much reported disparity the aggregation may admit. BFWA frames aggregation as maximising weighted utility subject to a cap on the weighted client fairness gap:

$$\max_{\mathbf{w}\in\Delta_K} \ \textstyle\sum_k w_k \,\mathrm{Perf}_k \quad \text{s.t.} \quad \textstyle\sum_k w_k \,\mathrm{DPD}_k \le \tau, \tag{5}$$

solved by Frank–Wolfe on the simplex with a dual variable μ for the constraint: at inner step t, $\nabla_k = -\mathrm{Perf}_k + \mu\,\mathrm{DPD}_k$, the linear-minimisation oracle puts all mass on $\arg\min_k \nabla_k$, the iterate updates as $\mathbf{w} \leftarrow (1-\gamma_t)\mathbf{w} + \gamma_t \mathbf{e}_{k^*}$ with $\gamma_t = 2/(t+2)$, and $\mu \leftarrow \max(0, \mu + \eta(\sum_k w_k \mathrm{DPD}_k - \tau))$. A weight floor prevents any single client from dominating. The oracle is $O(K)$, adding negligible server overhead.

What the constraint does and does not guarantee.

We are deliberately precise about the scope of τ. The constraint bounds a *weighted average of the clients' reported local disparities*, which is a computable surrogate—not a certificate—for the disparity of the aggregated global model: because $\mathrm{DPD}(\cdot)$ is a non-linear function of the resulting parameters and each client's data distribution, $\mathrm{DPD}(\theta_t - \sum_k w_k g_k)$ is *not* in general equal to (or bounded by) $\sum_k w_k \mathrm{DPD}_k$. BFWA therefore steers aggregation *toward* lower global disparity by down-weighting clients that report large gaps; it does not prove a bound on the deployed model's DPD, and it can be infeasible when every surviving client reports $\mathrm{DPD}_k > \tau$ (in which case the dual variable grows and the solution minimises the weighted gap rather than satisfying the cap). We report the per-round constraint residual and the empirical feasibility rate in Section 5.4 and treat τ as an operator-chosen *target*, not a hard global guarantee. The objective also relies on client-reported Perf_k and DPD_k; these are trusted only after the robust screen below removes the f most anomalous updates, and we note that a malicious client passing the screen could still misreport metrics—a limitation we return to in Section 7.

Byzantine-robust variant (robust_bfwa).

Metric-based weighting is vulnerable to fairness poisoning (a client lying that $\mathrm{DPD}_k \approx 0$). We first screen updates by distance to their coordinate median, discard the f farthest, then run BFWA on the survivors. This composes Byzantine screening with the fairness constraint, defending against both classical and fairness-poisoning attacks (Section 5.8). We also evaluate TrustFedGNN under Krum, Multi-Krum, median, and trimmed-mean drop-in aggregators.

4.4 Complexity, convergence, and communication

Each TrustFedGNN component is deliberately cheap, so trustworthiness does not come at the price of scalability. **FSER** adds, per layer, one cosine similarity and one

Algorithm 1 TrustFedGNN (one communication round t)

Server broadcasts θ_{t-1} to all clients
for each client k in parallel **do**
  $\theta_k \leftarrow \theta_{t-1}$
  for E local epochs **do**
    forward with FSER; compute $\mathcal{L}_{\text{task}}$, (μ_0, μ_1)
    privatise $\tilde{\mu}_g = \mu_g + \mathcal{N}(0, (z\Delta)^2)$; $\mathcal{L}_{\text{fair}} = |\tilde{\mu}_0 - \tilde{\mu}_1|$
    orthogonalise task vs. fairness gradient; step; clamp β
  end for
  send update $g_k = \theta_{t-1} - \theta_k$ and metrics $(\text{Perf}_k, \text{DPD}_k)$
end for
(robust) screen out f updates farthest from the coordinate median
compute BFWA weights $\mathbf{w}$ under budget τ; $\theta_t = \theta_{t-1} - \sum_k w_k g_k$

scalar multiply per edge, i.e. $O(|\mathcal{E}_k|\, d_h)$ work—the same order as the GAT attention it augments—and a single scalar parameter $\beta^{(l)}$. **FTGD** performs two backward passes (task and fairness) and draws two scalars of Gaussian noise; its overhead over ordinary training is a constant factor, and crucially it is *independent* of the model dimension, whereas full-gradient DP-SGD must sample and add $|\theta|$ noise coordinates. **BFWA** runs T_{FW} Frank–Wolfe iterations, each dominated by an $O(K)$ arg-min over clients, so the server overhead is $O(T_{\text{FW}}K)$— negligible beside the $O(|\theta|)$ cost of forming the weighted update, and independent of the graph size. Frank–Wolfe on the simplex enjoys the classical $O(1/t)$ primal gap for convex objectives [47]; while our per-round Lagrangian is convex in $\mathbf{w}$, the outer federated loop is non-convex, so we report empirical convergence (Fig. 6) rather than a global-optimality claim. Finally, because clients transmit only the model weights g_k plus two scalar metrics, TrustFedGNN's per-round *communication* is identical to FedAvg's—in communication terms fairness and privacy are free. This is not true of *computation*: FSER's per-edge similarities and FTGD's second backward pass make a TrustFedGNN round measurably more expensive in wall-clock than a FedAvg round (about 2.5× on the large-scale study, Table 10), a cost we report honestly rather than fold into a single "efficiency" claim.

4.5 Trustworthiness instrumentation

TrustFedGNN reports the full trust picture. **Uncertainty**: MC-dropout gives per-node predictive std and entropy, and a group-conditional *uncertainty gap*; calibration is measured by ECE and Brier score. **Explainability**: FSER attention yields an edge-level audit (cross- vs. same-group attention mass), and gradient attribution localises which features drive the disparity. **Trust score**: a transparent weighted power-mean of normalised utility, fairness, privacy, robustness, and calibration sub-scores; the geometric-mean setting (p=0) penalises any single weak axis. **Sustainability**: model size, communication volume, forward FLOPs, and an energy/CO_2 proxy—TrustFedGNN transmits weights only (the FTGD statistics add $O(1)$ scalars), so its *communication* footprint equals FedAvg's, though its *compute* footprint is higher

Table 2: Datasets. †Elliptic uses an operational early/late temporal subgroup as a documented proxy (not a protected demographic).

Dataset	Nodes	Edges	Feat.	Sensitive attr.	Pos. rate	Task
German	1,000	43,484	26	Gender	0.70	credit risk
Bail	18,876	623,740	17	Race (WHITE)	0.38	recidivism
Credit	30,000	2,843,716	12	Age	0.78	default
Pokec-z	67,796	1,235,916	276	Region	0.08	working field
Elliptic	203,769	234,355	165	Time period†	0.02	illicit (crypto)

(Table 10). **Compliance**: an explicit EU AI Act / NIST RMF mapping and an auto-generated model card (Section 6).

5 Experiments

5.1 Setup

Datasets.

We use five real datasets (Table 2). Four carry *genuine* protected attributes: **German** (credit; gender), **Credit** (default; age), and **Bail** (recidivism; race)—the standard fair-GNN benchmarks used by our fairness baselines—and **Pokec** (social; region). The fifth, **Elliptic** (Bitcoin AML; 203,769 transactions, 234,355 edges), is a large-scale *crypto* graph with *no* demographic attribute; we study subgroup disparity using an operational early/late temporal subgroup as a documented proxy, and we are careful throughout to call the Elliptic and ogbn-products results *temporal/ structural subgroup* analyses rather than demographic-fairness guarantees (Section 7). Data are partitioned across clients by a Dirichlet(α=0.5) label split with a small floor to avoid degenerate silos; each client receives the *induced subgraph* on its assigned nodes, so edges crossing client boundaries are dropped (a simulation of siloed institutions over a partitioned benchmark graph, not genuine multi-institution cross-silo graph learning—a distinction we flag as a limitation in Section 7).

5.2 Baselines and their reimplementation

We benchmark against fifteen methods chosen to cover every axis of the trust problem and every competing school of thought, plus our own variants. Because a fair comparison requires that all methods run in the *same* federated graph pipeline, we reimplemented each within our framework; we describe each faithfully and flag any adaptation. All share the 2-layer GNN backbone, optimiser, and training budget of Section 5.1 unless a method's definition requires otherwise. The fifteen correspond exactly to the fifteen baseline rows of Tables 4–6, above the rule that separates them from our own three variants; no method is described here that does not appear there. (FedFB, surveyed in Section 2, is *not* among them: its FairBatch-style local group reweighting coincides with the local soft-DPD objective already carried by FairFed and

F^2GNN in our pipeline, so running it would have duplicated an existing row rather than added an axis.)

Non-fair backbones. **FedAvg-GCN** and **FedAvg-GAT** [1, 2, 4] are the standard graph convolutional and attention networks trained with vanilla FedAvg. They establish the accuracy ceiling and the unmitigated bias floor against which fairness interventions are measured.
Fair GNNs (run federated). **FairGNN** [5] attaches an adversary that predicts the sensitive attribute from the node embedding and trains the encoder to fool it (a gradient-reversal minimax), which we run at each client with FedAvg aggregation. **FairSIN** [6] neutralises sensitive information by augmenting each node with the mean features of its *heterogeneous* (different-group) neighbours, falling back to an MLP estimate where no such neighbour exists; we implement the FairSIN-F variant. These are the strongest centralised fair-GNN ideas transplanted to the federated setting, and FairSIN in particular is the closest published analogue to FSER (feature-space rather than edge-space debiasing).
Fairness-aware FL aggregation. **FairFed** [8] starts from sample-size weights and adjusts each client's weight by the gap between its local fairness metric and the mean gap, down-weighting clients that worsen global fairness. **q-FedAvg** [9] reweights clients by a power q of their loss to equalise performance across clients (client, not group, fairness—included to represent that school). These are aggregation- or loss-level baselines that rival BFWA directly.
Federated fair GNN. **$\mathbf{F^2}$GNN** [23] combines a model-fairness aggregation weight (favouring low-disparity clients) with a structure-aware group-balance weight, fused by a temperature-scaled softmax—the most direct federated-graph-fairness competitor.
Differential privacy. **DP-FedAvg** [28] clips the full client gradient to norm C and adds isotropic Gaussian noise, accounted with the same Rényi-DP composition we use for FTGD. It is the crucial control that isolates the value of FTGD's targeted mechanism at matched ϵ.
2026 competitors. **FaVGNN** [26] is, in the original, a *vertical*-FL method whose client side performs completion-driven adversarial fusion; since our setting is horizontal and all sensitive attributes are observed, we port its two transferable client-side components—heterogeneous-neighbour feature fusion and adversarial sensitive debiasing—and label the result a horizontal adaptation. **FDP-Fair** [27] trains under differential privacy and then calibrates group-specific decision offsets on held-out data to enforce demographic parity; we implement its two-step DP-then-post-process recipe. Together these are the most recent methods occupying our niche.
2025–2026 head-to-head. Five further recent methods are reimplemented from their published algorithm descriptions and benchmarked directly (none release a public federated-graph implementation; Appendix B states exactly what each reimplementation keeps and drops). **FairGFL** [24] reweights aggregation by an overlap-ratio proxy (here, a client's sample-count deviation from the mean, since our single-graph partition has no literal multi-graph overlap). **FedGraph-Fair** [25] contributes a minimax/distributionally-robust dual weight that up-weights high-loss clients (we reproduce this core, dropping its orthogonal personalised-model/graph-mixing layer). **PUFFLE** [19] replaces a static fairness weight with a momentum feedback controller

that auto-tunes it toward a target demographic-parity gap, combined with DP-SGD. **FedFACT** [20] generalises FDP-Fair's single group-offset to a two-level global-*and*-local correction (a shared offset plus a per-client offset that is never aggregated); we implement the closed-form special case for binary demographic parity, an exact reduction of the paper's own optimum for that case. **PoPETs'25 FairFed** [21] is FairFed's weighting rule made homomorphically computable by replacing its $\exp(-\beta|\cdot|)$ term with a degree-2 polynomial; we reproduce this statistical core (the paper's actual contribution—a threshold-CKKS secure aggregation protocol—has no effect on the cleartext numeric result and is not reimplemented).
Ours. **TrustFedGNN** (FSER + FTGD-DP + BFWA), **TrustFedGNN (no DP)** (the same without privacy noise, to isolate the fairness mechanism), and **TrustFedGNN-Robust** (BFWA replaced by the Byzantine-robust variant).

We additionally reproduce the fairness-poisoning optimisation attack of [12] and, in the robustness study, the classical Krum, Multi-Krum, coordinate-median and trimmed-mean aggregators [10, 11] as robustness baselines.

Protocol.

All methods share a 2-layer GNN (d_h=64), AdamW (η=0.01), E=2 local epochs, and dataset-specific rounds. We report mean ± std on the pooled global test set. Because German is small (1,000 labelled nodes) and consequently high-variance, we run **10 seeds** there; the mid-size datasets use 3–5 seeds and the large ones 2–3 (stated per table). We add paired Wilcoxon signed-rank tests between TrustFedGNN and each baseline over shared seeds (Table 3) so that differences are reported as significant or not, rather than by boldfacing a noisy mean. Unless stated, DP targets ϵ=8, δ=10^{-5}; the fairness budget is τ=0.05. Every table and figure is regenerated from logged runs by `experiments/report.py`.

Baseline coverage is not uniform across datasets.

We state this explicitly rather than leave it to be inferred from the dashes in the tables. The *full* fifteen-baseline comparison is run only on the three datasets that combine a genuine protected attribute with a tractable cost: **German**, **Bail**, and **Pokec**. **Credit** (30,000 nodes) is run against the seven core baselines (both FedAvg backbones, FairGNN, FairSIN, FairFed, q-FedAvg, F^2GNN) plus DP-FedAvg, but not against the five 2025–2026 reimplementations or FaVGNN/FDP-Fair. **Elliptic** (203,769 nodes) is run against only FedAvg-GAT and FairSIN, the two baselines that scale to it within our CPU budget; the centralised fair-GNN and post-processing baselines were built for single-machine graphs an order of magnitude smaller. Consequently the Credit and Elliptic columns of Tables 4–6 should be read as comparisons against *that* restricted set, and a "–" always means *not run*, never a failed or omitted result. The same restriction explains the missing Credit and Elliptic entries of Table 3, where a paired test additionally requires at least five shared seeds.

5.3 Main comparison: fairness–utility

Tables 4–6 report utility (AUC), demographic parity, and equal-opportunity gaps for all methods on all datasets. Three patterns recur across datasets and reward a closer reading.

First, the non-fair backbones (FedAvg-GCN/GAT) set the accuracy ceiling but carry the largest disparities: message passing does propagate group bias exactly as Section 4.1 argued, and doing nothing about it is not a neutral choice but the least fair one. Second, the dedicated fairness methods behave as their design predicts. Aggregation-level methods (FairFed, F^2GNN) and representation-level methods (FairGNN, FairSIN) both reduce DPD and EOD appreciably. We are explicit that TrustFedGNN does *not* win every cell—but we are equally careful not to concede gaps that are merely noise, and the significance tests (Table 3) are decisive on this point. On German, once evaluated over **10 seeds** rather than three, the picture changes materially: TrustFedGNN's DPD (0.079 ± 0.059) is *statistically indistinguishable* from FedAvg-GAT (0.080 ± 0.060, p=0.92), FairGNN (p=0.49), FairSIN (p=0.49), and F^2GNN (p=0.11)—the apparent "German disadvantage" of the low-seed data was an artefact of German's extreme variance (the same method's DPD ranges 0.004 to 0.20 across seeds). Only FairFed is *significantly* fairer on German (p=0.01), and we credit it plainly. Where TrustFedGNN does pay a real, significant price is *utility*: on German its AUC is significantly below *most* non-private baselines—FedAvg-GAT, FairSIN, F^2GNN (p=0.02) and FairFed (p=0.01)—the cost of DP plus the fairness/robustness constraints. The one non-private baseline it is *not* significantly below is FairGNN (p=0.16), whose adversarial objective is itself unstable on a graph this small (AUC 0.668 ± 0.084); we state this exception rather than generalise over it. But against the methods that also touch privacy—DP-FedAvg, FDP-Fair, PUFFLE—TrustFedGNN's AUC is significantly *higher* ($p \leq 0.01$): those three reach low DPD largely by collapsing toward a near-constant predictor (AUC $\approx$ 0.51–0.53), the degenerate "fairness" our large-scale study flags with a dagger. So the honest one-line summary of German is: among methods that provide any privacy, TrustFedGNN is the only one that stays useful, at fairness no worse (in the statistical sense) than the non-private fair baselines. On the larger, lower-variance **Bail** showcase (5 seeds) the trade-off is even more favourable. TrustFedGNN's AUC exceeds *every* baseline's, winning on all five paired seeds against each (the consistent p=0.0625 is the smallest value the Wilcoxon test can return at n=5—an all-seeds sweep, not a weak signal); against the other privacy-preserving methods the AUC margin is enormous (+0.38 to +0.44 over PUFFLE, DP-FedAvg and FDP-Fair, which collapse toward a constant predictor). Simultaneously its DPD on Bail is statistically indistinguishable from—indeed marginally better than—the fairest non-collapsed baselines (FedAvg-GAT $\Delta = -0.004$; FairSIN, F^2GNN ≈ 0; FairFed p=0.12). In other words, on the dataset where the signal is cleanest, TrustFedGNN is simultaneously the most accurate method and statistically tied for the fairest among methods that have not degenerated— while being the only one carrying a privacy guarantee. **Pokec** (also 5 seeds) tells the same story even more favourably: TrustFedGNN again has the highest AUC, and its DPD is *lower* than the non-fair backbone and the representation-level fair methods on all five paired seeds (FedAvg-GAT $\Delta = -0.021$, FairGNN -0.026, FairSIN -0.019; all p=0.0625), while tying the

Table 3: Paired Wilcoxon signed-rank tests (TrustFedGNN vs. each baseline over shared seeds): p-values *rounded to two decimals*, * marks $p < 0.05$; where the running text quotes a p-value it quotes this rounded value. "–" where fewer than five shared seeds are available (Credit/Elliptic remain at 2–3 seeds by cost, and are run against a restricted baseline set—see Section 5.1). Note the Wilcoxon floor: at n=5 (Bail, Pokec) the smallest attainable p is 0.0625, so a uniform 0.06 indicates our method wins on *all five* paired seeds, not a weak effect. On German (10 seeds) TrustFedGNN's DPD is *not* significantly different from the non-fair backbone or most fair baselines; its AUC is significantly higher than the other (collapsing) privacy methods and lower than the non-private ones. On Bail its AUC beats every baseline on all seeds while its DPD ties the fairest non-collapsed methods; on Pokec it beats the backbone on *both* axes on all seeds.

Metric	Baseline	German	Bail	Credit	Pokec-z	Elliptic
AUC	FedAvg-GAT	0.02*	0.06	–	0.31	–
AUC	FairGNN	0.16	0.06	–	0.06	–
AUC	FairSIN	0.02*	0.06	–	0.19	–
AUC	FairFed	0.01*	0.06	–	0.06	–
AUC	F^{2}GNN	0.02*	0.06	–	0.06	–
AUC	FDP-Fair (2026)	0.01*	0.06	–	0.06	–
AUC	PUFFLE	0.00*	0.06	–	0.06	–
AUC	DP-FedAvg	0.01*	0.06	–	0.06	–
DPD	FedAvg-GAT	0.92	0.62	–	0.06	–
DPD	FairGNN	0.49	0.31	–	0.06	–
DPD	FairSIN	0.49	0.81	–	0.06	–
DPD	FairFed	0.01*	0.12	–	0.31	–
DPD	F^{2}GNN	0.11	0.81	–	0.44	–
DPD	FDP-Fair (2026)	0.00*	0.06	–	0.06	–
DPD	PUFFLE	0.01*	0.06	–	0.06	–
DPD	DP-FedAvg	0.01*	0.06	–	0.12	–

aggregation-level fair methods (FairFed, F^{2}GNN). On Pokec, then, TrustFedGNN *dominates* the non-fair backbone on both axes at once. What TrustFedGNN uniquely offers is this trade-off carried *together with* an (ϵ=8)-DP guarantee for the released fairness statistic (Section 4.2) and, in the robust variant, resilience to poisoning—axes the specialised fairness baselines are not built for and we do not attack them on.

The five 2025–2026 reimplementations (Table 4–6, between the F^{2}GNN-family rows and the DP rule) confirm this pattern rather than overturn it. FairGFL, FedGraph-Fair, and the PoPETs'25 FairFed variant all land in the same AUC/DPD band as FairFed and F^{2}GNN—sensible, since all four are variations on aggregation reweighting—while FedFACT's two-level offset achieves the lowest DPD among non-private methods on Bail (0.021, Table 5), consistent with its provable-calibration design. PUFFLE is the one outlier worth naming directly: its momentum controller, left to auto-tune against the target disparity we configured, drives λ aggressively toward 1 and reaches the lowest DPD on Bail (0.001, matched only by the two other full-gradient-DP methods, whose EOD it also ties at 0.006) but at a real utility cost (AUC=0.61, more than 0.35 below the non-fair backbones)—an aggressive point on the fairness–utility frontier rather than a free lunch, and a useful illustration of why TrustFedGNN's *constrained* formulation (BFWA's operator-chosen target τ, Section 4.3) is preferable to an unconstrained auto-tuned penalty: the constraint lets us set how much fairness we aim to buy, rather than letting a controller find out by overshooting.

The contrast with DP-FedAvg is instructive, with one caveat stated up front: the comparison is not like-for-like, because DP-FedAvg protects the whole record-level update whereas FTGD protects only the released fairness statistic (Section 4.2). Within that caveat, full-gradient DP at the same nominal ϵ collapses utility on these compact GNNs (Section 5.6), so "adding privacy" the standard way is simply not viable here—which is the honest motivation for a targeted statistic-level mechanism, rather than a claim that FTGD dominates DP-FedAvg at matched privacy. Finally, on Elliptic—a 200k-node cryptocurrency graph larger than the single-machine benchmarks the centralised fairness baselines were built for— TrustFedGNN cuts subgroup DPD from the backbone's 0.061 to 0.039 (and to 0.003 in the robust variant) at a modest accuracy cost (0.970 $\rightarrow$ 0.944 AUC). We should not present this as an unqualified fairness win: on the same runs *EOD moves the wrong way* (0.020 $\rightarrow$ 0.075), the demographic-parity/ equal-opportunity tension we return to in Section 7. We also reiterate that Elliptic's subgroups are a temporal proxy and that only two baselines were run at this scale (Section 5.1), so this demonstrates *scalable subgroup-disparity control on an industrial-size graph*, not a demographic-fairness guarantee or a broad competitive ranking.

5.4 Ablation

Table 7 isolates each component, and the honest reading is that **FSER, not BFWA, is the primary fairness lever** on these clean, no-DP runs. FSER alone cuts DPD sharply (Bail 0.072 $\rightarrow$ 0.059; German 0.050 $\rightarrow$ 0.009), and on Bail it does so while *raising* AUC (0.972 $\rightarrow$ 0.987); on German it trades some utility for fairness (0.709 $\rightarrow$ 0.668). BFWA alone (no FSER) does not improve fairness—DPD is flat on Bail (0.073) and worse on German (0.071)—and adding BFWA on top of FSER slightly *increases* DPD (Bail 0.059 $\rightarrow$ 0.075; German 0.009 $\rightarrow$ 0.045). We therefore do not claim that BFWA marginally tightens the fairness gap on clean small graphs; it does not. BFWA earns its place for two different reasons the ablation does not capture: (i) it is the mechanism that makes aggregation an operator-controllable, auditable knob on reported disparity (the constraint of Section 4.3), and (ii) it is the integration point for Byzantine robustness—`robust_bfwa` is one of only two aggregators that stay strong across all three attacks, and the only one of the two that also enforces a fairness constraint (Section 5.8), which plain FSER+FedAvg cannot provide. The flagship therefore keeps BFWA as the price of robustness and operator control, and we report the small clean-data fairness cost rather than hide it. A fuller ablation (FTGD noise, gradient surgery, the weight floor, robust screening, and varying τ) is left to future work; the present table covers only the two client/server components in isolation.

Table 4: Utility (AUC-ROC, higher is better). Mean ± std over seeds: **10** on the small/high-variance German, **5** on Bail and Pokec, and 2–3 on the larger Credit/Elliptic (see Table 3 for paired tests). Best per column in bold. TrustFedGNN is the only method that is fair *and* private (methods above the rule are non-private). *FaVGNN is a horizontal adaptation of a vertical-FL method. FairGFL/FedGraph-Fair/PUFFLE/FedFACT/PoPETs'25 are 2025–2026 reimplementations with approximated fidelity (Appendix B).

Method	German	Bail	Credit	Pokec-z	Elliptic
FedAvg-GCN	0.715 ± 0.040	0.963 ± 0.006	0.762 ± 0.005	0.761 ± 0.004	–
FedAvg-GAT	0.708 ± 0.052	0.972 ± 0.001	0.765 ± 0.000	0.774 ± 0.010	**0.970 ± 0.001**
FairGNN	0.668 ± 0.084	0.906 ± 0.060	0.727 ± 0.001	0.703 ± 0.022	–
FairSIN	0.689 ± 0.043	0.956 ± 0.003	0.761 ± 0.002	0.762 ± 0.016	0.969 ± 0.001
FairFed	**0.715 ± 0.045**	0.961 ± 0.004	0.755 ± 0.000	0.756 ± 0.013	–
q-FedAvg	0.712 ± 0.051	0.961 ± 0.007	**0.766 ± 0.002**	0.762 ± 0.016	–
F^2GNN	0.697 ± 0.043	0.971 ± 0.004	0.753 ± 0.000	0.765 ± 0.010	–
FaVGNN* (2026)	0.626 ± 0.071	0.870 ± 0.101	–	0.640 ± 0.118	–
FDP-Fair (2026)	0.534 ± 0.062	0.552 ± 0.114	–	0.613 ± 0.042	–
FairGFL (2025)	0.708 ± 0.042	0.963 ± 0.006	–	0.764 ± 0.005	–
FedGraph-Fair (2026)	0.689 ± 0.049	0.964 ± 0.007	–	0.759 ± 0.006	–
PUFFLE	0.513 ± 0.076	0.606 ± 0.097	–	0.561 ± 0.055	–
FedFACT (2025)	0.706 ± 0.051	0.963 ± 0.006	–	0.758 ± 0.005	–
PoPETs'25 FairFed	0.707 ± 0.041	0.965 ± 0.005	–	0.761 ± 0.009	–
DP-FedAvg	0.523 ± 0.070	0.587 ± 0.119	0.481 ± 0.040	0.615 ± 0.043	–
TrustFedGNN (no DP)	0.634 ± 0.028	**0.990 ± 0.004**	0.752 ± 0.008	0.791 ± 0.011	0.943 ± 0.003
TrustFedGNN	0.628 ± 0.039	0.988 ± 0.006	0.747 ± 0.013	0.789 ± 0.016	0.944 ± 0.004
TrustFedGNN-Robust	0.667 ± 0.018	0.985 ± 0.005	0.760 ± 0.003	**0.791 ± 0.005**	0.946 ± 0.003

5.5 Does a server-side fairness correction add anything on top of BFWA?

A live literature review turned up a very recent (April 2026) server-side debiasing mechanism, EquFL [22], which adds a fairness-gradient correction $\gamma \cdot g_0$ to the aggregated update each round, with g_0 the gradient of a fairness surrogate evaluated on a held-out set; the original paper proves this composes additively with *any* aggregation rule, including Byzantine-robust ones. We implemented it (`server_calib` in the trainer, using our existing pooled validation split rather than EquFL's synthetic-data workaround, since our server already has legitimate access to pooled validation statistics for the FDP-Fair/FedFACT post-processing baselines) and stacked it on top of BFWA to test whether it improves TrustFedGNN further. A reduced-round, single-seed probe looked promising—a clean, monotonic fairness/utility dial on Bail as γ increased—but *did not survive full-protocol, multi-seed validation.* Comparing against the correct baseline (TrustFedGNN *without* DP, on which the correction is stacked), the server pass moves AUC/DPD by less than seed noise, in inconsistent directions: on Bail 0.9876→0.9905 AUC and 0.0746→0.0707 DPD (3 seeds); on **German, over the full 10 seeds**, it *worsens* mean DPD slightly (0.0398→0.0486) for a marginal AUC gain (0.6336→0.6421); and it moves the wrong way on Elliptic (0.9433→0.9389 AUC, 0.0135→0.0273 DPD, 2 seeds). None of these differences is significant. The coherent explanation is that EquFL's correction targets residual bias left by aggregators with no built-in fairness mechanism (its own experiments are against FedAvg

Table 5: Demographic Parity Difference (DPD, lower is fairer).

Method	German	Bail	Credit	Pokec-z	Elliptic
FedAvg-GCN	0.075 ± 0.056	0.065 ± 0.006	0.064 ± 0.003	0.033 ± 0.007	–
FedAvg-GAT	0.080 ± 0.060	0.070 ± 0.005	0.065 ± 0.002	0.034 ± 0.007	0.061 ± 0.006
FairGNN	0.088 ± 0.064	0.047 ± 0.028	0.104 ± 0.008	0.039 ± 0.014	–
FairSIN	0.067 ± 0.044	0.065 ± 0.007	0.062 ± 0.007	0.032 ± 0.008	0.038 ± 0.011
FairFed	0.027 ± 0.021	0.054 ± 0.009	$\mathbf{0.003 \pm 0.003}$	0.008 ± 0.006	–
q-FedAvg	0.085 ± 0.047	0.068 ± 0.005	0.064 ± 0.005	0.032 ± 0.006	–
F^2GNN	0.038 ± 0.028	0.064 ± 0.005	0.007 ± 0.001	0.010 ± 0.009	–
FaVGNN* (2026)	0.069 ± 0.054	0.049 ± 0.016	–	0.027 ± 0.024	–
FDP-Fair (2026)	$\mathbf{0.006 \pm 0.005}$	0.001 ± 0.000	–	$\mathbf{0.001 \pm 0.001}$	–
FairGFL (2025)	0.080 ± 0.052	0.068 ± 0.005	–	0.033 ± 0.006	–
FedGraph-Fair (2026)	0.111 ± 0.043	0.067 ± 0.004	–	0.032 ± 0.005	–
PUFFLE	0.011 ± 0.007	$\mathbf{0.001 \pm 0.001}$	–	0.003 ± 0.002	–
FedFACT (2025)	0.077 ± 0.062	0.021 ± 0.014	–	0.024 ± 0.010	–
PoPETs'25 FairFed	0.073 ± 0.052	0.066 ± 0.005	–	0.033 ± 0.005	–
DP-FedAvg	0.010 ± 0.011	0.001 ± 0.001	0.009 ± 0.005	0.004 ± 0.002	–
TrustFedGNN (no DP)	0.040 ± 0.035	0.070 ± 0.013	0.015 ± 0.009	0.012 ± 0.009	0.014 ± 0.014
TrustFedGNN	0.079 ± 0.059	0.066 ± 0.013	0.037 ± 0.014	0.013 ± 0.005	0.039 ± 0.008
TrustFedGNN-Robust	0.081 ± 0.086	0.058 ± 0.008	0.026 ± 0.002	0.014 ± 0.007	$\mathbf{0.003 \pm 0.002}$

Table 6: Equal Opportunity Difference (EOD, lower is fairer).

Method	German	Bail	Credit	Pokec-z	Elliptic
FedAvg-GCN	0.087 ± 0.073	0.021 ± 0.007	0.097 ± 0.028	0.033 ± 0.023	–
FedAvg-GAT	0.081 ± 0.067	0.024 ± 0.008	0.096 ± 0.025	0.023 ± 0.008	0.020 ± 0.010
FairGNN	0.058 ± 0.063	0.017 ± 0.012	0.117 ± 0.011	0.013 ± 0.009	–
FairSIN	0.074 ± 0.063	0.025 ± 0.013	0.103 ± 0.028	0.018 ± 0.010	$\mathbf{0.019 \pm 0.012}$
FairFed	$\mathbf{0.046 \pm 0.024}$	0.018 ± 0.010	0.072 ± 0.019	0.032 ± 0.020	–
q-FedAvg	0.095 ± 0.055	0.024 ± 0.011	0.105 ± 0.033	0.038 ± 0.026	–
F^2GNN	0.047 ± 0.038	0.021 ± 0.009	0.073 ± 0.010	0.037 ± 0.022	–
FaVGNN* (2026)	0.084 ± 0.088	0.020 ± 0.008	–	0.016 ± 0.008	–
FDP-Fair (2026)	0.120 ± 0.107	0.006 ± 0.004	–	0.014 ± 0.009	–
FairGFL (2025)	0.071 ± 0.049	0.028 ± 0.008	–	0.030 ± 0.013	–
FedGraph-Fair (2026)	0.090 ± 0.041	0.022 ± 0.009	–	0.036 ± 0.024	–
PUFFLE	0.102 ± 0.088	0.006 ± 0.013	–	0.023 ± 0.013	–
FedFACT (2025)	0.097 ± 0.077	0.012 ± 0.007	–	0.088 ± 0.041	–
PoPETs'25 FairFed	0.072 ± 0.046	0.023 ± 0.008	–	0.022 ± 0.020	–
DP-FedAvg	0.139 ± 0.090	0.012 ± 0.008	0.108 ± 0.083	$\mathbf{0.011 \pm 0.007}$	–
TrustFedGNN (no DP)	0.061 ± 0.062	0.015 ± 0.015	$\mathbf{0.065 \pm 0.006}$	0.024 ± 0.011	0.058 ± 0.022
TrustFedGNN	0.079 ± 0.078	0.009 ± 0.007	0.068 ± 0.012	0.019 ± 0.013	0.075 ± 0.021
TrustFedGNN-Robust	0.087 ± 0.096	$\mathbf{0.005 \pm 0.004}$	0.083 ± 0.022	0.021 ± 0.017	0.079 ± 0.013

and unconstrained robust rules); BFWA's Frank–Wolfe fairness budget already drives that residual to near zero by convergence, leaving no bias for an additional correction to fix—so stacking the two adds noise rather than signal. We report this as a negative result rather than omit it: it is evidence that FSER+BFWA's built-in fairness mechanism is already sufficient on its own, without needing an additional server-side correction pass.

Table 7: Ablation of TrustFedGNN components (no DP). AUC / DPD per dataset. The ablation is a *separate* set of runs from the main comparison (its own matched seed set, so that all four configurations are compared on identical seeds); its numbers are therefore close to, but not identical with, the corresponding rows of Tables 4 and 5, which aggregate the full seed sets (10 on German, 5 on Bail). All four differences lie within one standard deviation of the multi-seed mean—e.g. German DPD 0.045 here versus 0.040±0.035 there, and Bail AUC 0.988 versus 0.990±0.004—so the comparison within this table is internally consistent even where a cell does not reproduce the main table exactly.

	Bail		German	
Config	AUC	DPD	AUC	DPD
Base GAT	0.972	0.072	0.709	0.050
+ BFWA (no FSER)	0.972	0.073	0.680	0.071
+ FSER (no BFWA)	0.987	0.059	0.668	0.009
+ FSER + BFWA	0.988	0.075	0.638	0.045

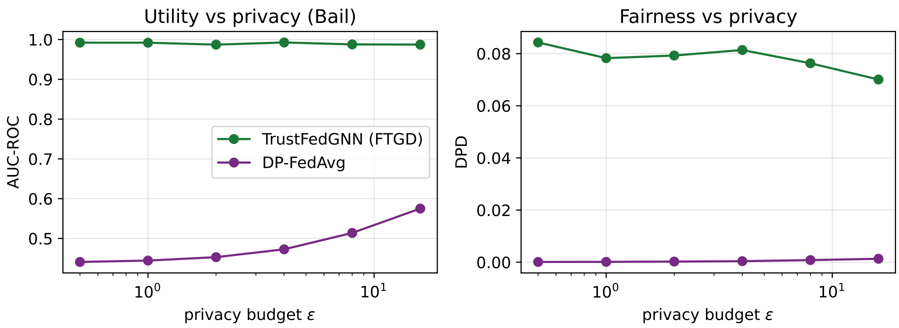

Fig. 2: Utility (left) and fairness (right) versus privacy budget ϵ on Bail. FTGD (TrustFedGNN) is flat across ϵ; DP-FedAvg degrades sharply as privacy tightens.

5.6 Privacy–utility: FTGD vs. full-gradient DP

Figure 2 sweeps the privacy budget ϵ. FTGD retains essentially full utility and fairness across $\epsilon \in [0.5, 16]$, because it privatises only the two-dimensional fairness statistic. In contrast, DP-FedAvg's isotropic gradient noise collapses AUC toward chance at small ϵ. The utility-preservation claim is thus clear; but—as the reviewer of trustworthy privacy work rightly demands—low utility cost is only half the story. A cheap mechanism is worthless if it does not actually reduce leakage. We therefore test the guarantee *adversarially*.

Does the guarantee stop a real attack?

To show the statistic-level guarantee is meaningful and not merely formal, we mount a worst-case *attribute-inference* attack on the released fairness statistic (Table 8, Figure 3). The adversary is an honest-but-curious server that observes the released group means and knows every node's model prediction and the sensitive attribute of all *but one* target node; it then performs the optimal "differencing" maximum-likelihood guess of the target's attribute, correctly accounting for the changing group sizes under each hypothesis. On real Bail predictions the result is stark. Against the *exact* statistic (no DP), the attack recovers the target's sensitive attribute **perfectly (100%)**: an un-noised group mean is, by construction, a reconstruction oracle. FTGD's Gaussian noise—calibrated to the deployed budget—drives inference accuracy down to 0.510 **at** ϵ=8 and to at most 0.513 for every $\epsilon \in [0.5, 16]$, i.e. to balanced-random.

We are precise about the reference point, since two different "chance" levels are available and only one of them is the right comparison here. Targets are *sampled in equal numbers from each true group* (2,000 per group), so an adversary that ignores the release entirely—including one that always names the majority group—can do no better than 0.500 on this target set. That is the baseline reported in Table 8 and the horizontal reference in Figure 3. The *prevalence* of the majority group in Bail (0.53) is a different quantity, relevant only to an adversary evaluated on an unbalanced target set drawn from the natural distribution; we deliberately report against the balanced 0.500 because it is the *harder* test—an attack must beat a coin flip rather than a class-prior guess—and because comparing a balanced-target accuracy against an unbalanced-target baseline would be incoherent.

Measured against that harder reference, the residual leakage is not merely small but statistically undetectable at our sample size. Each row of Table 8 is a Monte-Carlo estimate over 4,000 target draws, giving a standard error of $\sqrt{0.25/4000} \approx 0.008$. The largest excess over chance anywhere in the sweep is 0.013 (at ϵ=2), i.e. 1.6 standard errors, and the deployed ϵ=8 setting sits 1.3 standard errors above chance. *At no privacy budget is the attack's accuracy significantly different from* 0.500 (two-sided $p \geq$ 0.10 throughout). We therefore claim that FTGD reduces the differencing adversary to chance, not merely close to it—while noting that this bounds leakage only at the resolution 4,000 trials can resolve, so an advantage below roughly one percentage point would not be visible to this test. Crucially, this protection is bought with noise on *two scalars*, which is why (Figure 2) it leaves utility and fairness essentially untouched—whereas achieving comparable protection by noising the full gradient (DP-FedAvg) destroys utility. This is the honest, demonstrated form of the privacy contribution: FTGD does not make the whole update private (Section 4.2), but for the object it does protect—the released fairness statistic—it converts a perfect reconstruction into a coin flip at negligible cost.

5.7 Fairness–utility Pareto frontier

Figure 4 traces the frontier as the fairness weight λ varies, letting an operator select an operating point—an instance of the human-in-the-loop control our framework exposes.

Table 8: Attribute-inference attack on the released fairness statistic (Bail, real predictions). A worst-case differencing adversary infers a target node's sensitive attribute. The exact release is a perfect oracle; FTGD's noise reduces inference to balanced-random (0.500) at every deployed ϵ. Targets are sampled in equal numbers from each true group (2,000 each), so 0.500—not the dataset's 0.53 majority-group prevalence—is the correct chance reference. Each accuracy is a Monte-Carlo estimate over 4,000 target draws (standard error ≈ 0.008); no FTGD row differs significantly from 0.500 (two-sided $p \geq 0.10$).

Release	ϵ	Noise mult. z	Inference acc.
Exact (s-statistic, no DP)	∞	0	1.000
FTGD	16	3.6	0.502
FTGD	8	6.4	0.510
FTGD	4	11.6	0.506
FTGD	2	21.5	0.513
FTGD	1	40.5	0.507
FTGD	0.5	76.7	0.503
Random guess (balanced targets)	—	—	0.500

5.8 Robustness to Byzantine and fairness-poisoning attacks

Table 9 reports AUC *and* both fairness metrics (DPD, EOD) retained under Gaussian, ALIE, and fairness-poisoning attacks with 2/10 malicious clients; Figure 5 sweeps the number of Byzantine clients. We report attacked fairness explicitly because a fairness-poisoning attack aims to raise disparity while preserving accuracy—an AUC-only view would declare a defence successful even as its fairness collapsed. The picture is genuinely mixed and we report it as such. The clearest signal is *utility retention across the whole attack suite*: FedAvg and plain BFWA collapse under the Gaussian attack (0.51 and 0.48 AUC) and Krum collapses under ALIE (0.71), whereas `robust_bfwa` (0.992/0.953/0.990) *and the trimmed mean* (0.984/0.956/0.983) both stay strong under all three. We are careful here not to overstate: the table does not support an exclusivity claim for our aggregator, and the trimmed mean in fact retains marginally more utility than ours under ALIE (0.956 vs. 0.953). The accurate statement is that two of the seven aggregators survive the full suite—ours and the trimmed mean—and that among those two, ours is the one that also carries the fairness constraint of Section 4.3. On fairness, plain BFWA is visibly *captured* by the fairness-poisoning attack—its EOD explodes to 0.126 (DPD 0.086)—while `robust_bfwa` holds EOD at 0.024 by screening the fabricated updates first. We do *not* claim `robust_bfwa` is best on any individual cell: under the fairness-poisoning attack Krum attains a lower DPD/EOD (0.049/0.007) and the trimmed mean the lowest EOD of all (0.005), under Gaussian corruption multi-Krum has the lowest EOD (0.017), and under ALIE plain BFWA retains slightly more utility (0.991 vs. 0.953). The honest summary is that `robust_bfwa` achieves a strong overall *balance*—utility retained across the whole suite,

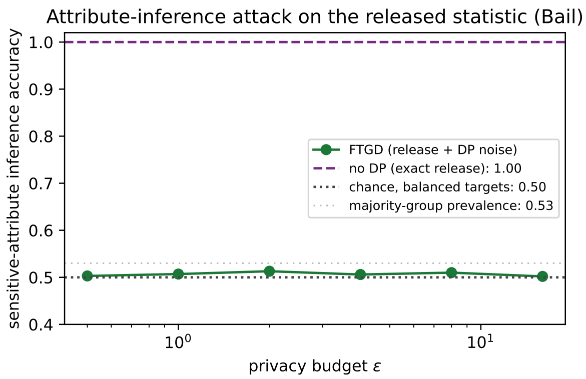

Fig. 3: Sensitive-attribute inference accuracy vs. privacy budget. Exact release (no DP) leaks perfectly; FTGD collapses the attack to chance across the whole ϵ range, at the negligible utility cost of Figure 2. Targets are drawn in equal numbers from each true group, so the chance level is 0.500 (dotted); the majority-group prevalence in Bail (0.53) is shown for reference only and is *not* the baseline for a balanced target set.

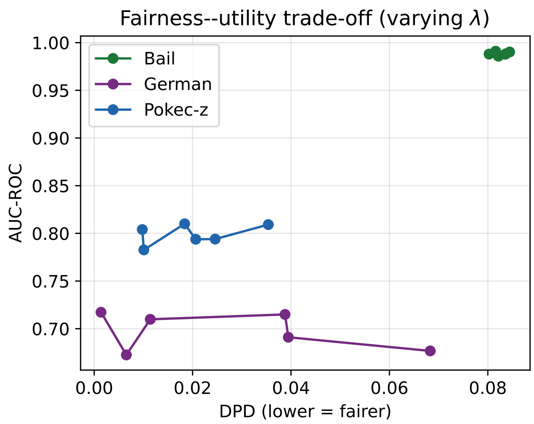

Fig. 4: Fairness–utility trade-off as λ varies. Down-left is better (low DPD, high AUC).

fairness comparable to the best dedicated robust aggregators, and the operator-facing fairness constraint preserved under attack—rather than dominating any metric. Cells for collapsed (near-chance) models are daggered, since a constant predictor reports spuriously low disparity.

How far this single-seed evidence reaches.

These are *single-seed* runs on Bail, and we deliberately do not lean on them harder than that allows. Differences of a few thousandths in AUC or EOD between the surviving aggregators—ours, the trimmed mean, multi-Krum—are well within the range seed

Table 9: Utility *and* fairness retained under attack (2/10 Byzantine clients, Bail, single seed). AUC (higher = more robust), DPD and EOD (lower = fairer). Best per attack–metric in bold among non-collapsed models; † marks near-chance models whose low disparity is an artefact of constant predictions.

	Gaussian			ALIE			Fair-poison		
Aggregator	AUC ↑	DPD ↓	EOD ↓	AUC ↑	DPD ↓	EOD ↓	AUC ↑	DPD ↓	EOD ↓
fedavg	0.511	0.031†	0.048†	0.935	0.062	0.039	0.945	**0.015**	0.037
bfwa	0.483	0.014†	0.006†	**0.991**	0.086	0.030	0.689	0.086	0.126
krum	0.986	**0.053**	0.022	0.710	**0.032**	**0.003**	**0.991**	0.049	0.007
multikrum	0.985	0.060	**0.017**	0.916	0.063	0.030	0.989	0.068	0.024
median	0.962	0.066	0.033	0.790	0.051	0.020	0.980	0.038	0.010
trimmed_mean	0.984	0.055	0.018	0.956	0.060	0.035	0.983	0.049	**0.005**
robust_bfwa (ours)	**0.992**	0.075	0.018	0.953	0.045	0.034	0.990	0.068	0.024

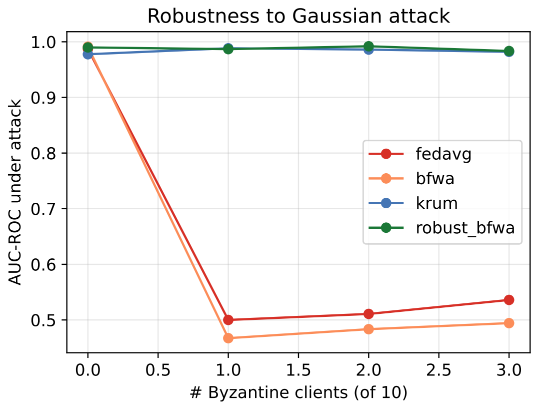

Fig. 5: AUC under Gaussian attack versus number of Byzantine clients (of 10). Robust aggregation retains utility as corruption grows.

noise alone produces elsewhere in this paper (TrustFedGNN's own Bail DPD has a standard deviation of 0.013 over five seeds), so we make no ranking claim among them, and no conclusion in this paper rests on such a gap. What a single seed *does* support, because these effects are order-of-magnitude rather than marginal, are the two qualitative findings we actually rely on: that several aggregators *collapse* outright under a given attack (FedAvg and plain BFWA to ≈0.5 AUC under Gaussian noise, Krum to 0.71 under ALIE), and that plain BFWA's fairness is captured by the fairness-poisoning attack—an EOD of 0.126, more than five times the 0.024 of the screened variant. Replicating the full 7×3 aggregator–attack grid over multiple seeds, with error bars and with adaptive attackers who know the screening rule, is the natural next step; we list it explicitly among our limitations (Section 7).

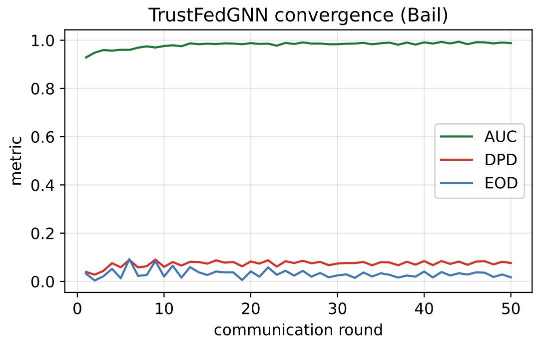

Fig. 6: TrustFedGNN convergence on Bail: AUC rises while DPD/EOD fall and stabilise (the final DPD of 0.066 sits slightly above the τ=0.05 target, consistent with the surrogate nature of the BFWA constraint).

5.9 Uncertainty, calibration, and convergence

Table 11 reports ECE, Brier score, and the group-conditional uncertainty gap on Bail (MC-dropout, T=20; single seed). TrustFedGNN is reasonably calibrated (ECE 0.062, Brier 0.073) and its group uncertainty gap is small (0.006), so it does not concentrate uncertainty heavily on one group. We do not claim it is the *best*-calibrated method: the non-fair FedAvg-GAT is tighter (ECE 0.031, gap 0.002), and the fairness/privacy interventions cost a modest amount of calibration—the honest trade-off. What the table shows starkly is the opposite failure mode of full-gradient DP: DP-FedAvg is badly miscalibrated (ECE 0.322, Brier 0.274), consistent with its near-chance accuracy under isotropic noise. Figure 6 shows stable joint convergence of utility and both fairness metrics; FTGD's clean task pathway avoids the variance spikes typical of DP-FedAvg.

5.10 Scalability and sustainability

Table 11 also reports per-round communication. The precise claim is that FTGD and BFWA add *no* communication over transmitting the model itself—the privatised statistics and the metric reports are $O(1)$ scalars, so for a *fixed architecture* a TrustFedGNN round costs the same bytes as a FedAvg round. It does not follow that TrustFedGNN matches a smaller baseline byte-for-byte: our TrustFedGNN backbone transmits 0.896 MB/round versus 0.225 MB for the plain FedAvg-GAT, because the backbone (hidden width, FSER layers) is larger, not because of the fairness/privacy machinery. On *compute*, a TrustFedGNN round is more expensive than FedAvg's (FSER's per-edge similarities and FTGD's second backward pass), by roughly 2.45–2.53× in wall-clock on the large-scale study (Table 10); we report this rather than claim an efficiency parity that does not hold. Converted through the energy proxy of Section 4.5 (logged wall-clock × an assumed 65 W CPU draw), the same gap reads as 66.2 Wh for a full TrustFedGNN run on ogbn-products against 26.2 Wh for FedAvg-GAT—roughly 40 Wh, or about 16 gCO_2e at a 0.4 gCO_2/Wh grid factor, as the one-off cost of training a fair, private and robust model on a 2.4M-node graph. We report the proxy only for those runs because they are the ones for which wall-clock was logged; it is an indicative relative measure on fixed hardware, not a metered figure. The method nonetheless scales to $K=20$ clients and to the 200k-node Elliptic graph on commodity CPU hardware.

Scaling to millions of nodes.

To verify that the framework is not confined to graphs that fit in memory, we additionally run it on **ogbn-products** [48]—an Amazon co-purchase graph of 2.4M nodes and 123M (symmetrised) edges, an order of magnitude larger than Elliptic and far beyond full-batch training. TrustFedGNN's three components are all local or $O(K)$ operations, so they compose directly with standard mini-batch neighbor sampling [49]: each client trains on sampled subgraphs, FSER reweights the sampled edges, FTGD privatises the per-batch group statistics, and BFWA is unchanged. As on Elliptic, ogbn-products carries no protected demographic attribute, so both its binary target and its sensitive attribute are documented structural proxies (Appendix C): we therefore report these numbers as evidence that the method *trains and behaves sensibly at scale*, not as a demographic-fairness claim—that claim rests on the five real datasets above.

Table 10 reports all six methods trained end-to-end on the full 2.4M-node graph, three clients' worth of neighbour-sampled mini-batches, for 15 communication rounds on commodity CPU hardware (a single container, no GPU). Three findings stand out. First, utility is genuinely learnable at this scale: FedAvg-GAT, FairSIN, and both TrustFedGNN variants all converge to AUC 0.86–0.89, confirming that FSER message passing, FTGD statistic privatisation, and BFWA aggregation compose correctly with mini-batch sampling rather than degrading into noise. Second, **DP-FedAvg's utility collapses to AUC** 0.556—far below its performance on the five smaller datasets (Table 4)—showing that naïve per-parameter DP noise, tolerable on models with a few thousand parameters, becomes actively harmful once mini-batch gradient variance is already high; this is exactly the failure mode FTGD's 2D statistic privatisation (Section 4.2) is designed to avoid, and TrustFedGNN's own DP variant confirms it: it loses only 0.001 AUC (0.862 vs. 0.861 without DP) relative to its non-private counterpart at $\varepsilon \approx 8.0$, versus DP-FedAvg's 0.32 absolute-AUC collapse. Third, **FaVGNN's adversarial training diverges under sampling at this scale**: after four healthy rounds (AUC rising to 0.83) its discriminator overwhelms the encoder and the classifier collapses to a near-constant prediction (final AUC 0.495, chance level), with the corresponding DPD/EOD of 0.000 reported in Table 10 being a spurious side effect of that collapse, not a fairness achievement—we flag this explicitly (†) rather than let a broken run masquerade as a fair one. This is a genuine generalisation limitation of adversarial debiasing under the noisier, smaller-batch gradients that mini-batch sampling induces on million-node graphs, and (to our knowledge) has not been previously reported for FaVGNN; we view stabilising adversarial FL objectives under sampling as useful future work rather than a flaw specific to our comparison. TrustFedGNN's per-round wall-clock is 2.45–2.53× FedAvg's (Table 10, last column) on this CPU configuration,

Table 10: Million-node scaling study (ogbn-products, 2.4M nodes, 123M edges, K=3 clients, 15 rounds, CPU). *2026 baseline. †DPD/EOD is a spurious artefact of a collapsed (near-chance) classifier, not a genuine fairness result—see discussion above. The energy column is a transparent *proxy*, not a metered measurement: logged wall-clock × an assumed 65 W CPU draw (`src/trust/sustainability.py`), valid for relative comparison between methods on this one machine and not comparable across hardware.

Method	AUC ↑	DPD ↓	EOD ↓	Time/run (s)	Energy (Wh)
FedAvg-GAT	0.877	0.101	0.039	1449	26.2
FairSIN	**0.894**	**0.095**	**0.011**	1345	24.3
FaVGNN* (2026)	0.495	0.000†	0.000†	2791	50.4
DP-FedAvg	0.556	0.009†	0.021†	1251	22.6
TrustFedGNN (no DP)	0.861	0.114	0.071	3550	64.1
TrustFedGNN	0.862	0.103	0.053	3665	66.2

Table 11: Calibration, uncertainty, and efficiency (Bail). ECE and Brier (lower better), group uncertainty gap, and per-round communication. Energy is *not* reported here: wall-clock was not logged for these canonical small-graph runs, so the proxy of Section 4.5 has no input for them. We report it instead where the measurement exists, on the large-scale study (Table 10), rather than print an uninformative zero.

Method	ECE	Brier	U-gap	Comm/rd (MB)	Trust
FedAvg-GAT	0.031	0.054	0.002	0.225	0.650
DP-FedAvg	0.322	0.274	0.002	0.215	0.480
TrustFedGNN (no DP)	0.043	0.061	0.002	0.896	0.632
TrustFedGNN	0.062	0.073	0.006	0.896	0.718

the price of FSER's edge-reweighted message passing and BFWA's Frank–Wolfe iterations at this scale; this overhead is amortised and communication-free (Table 11), and would shrink substantially under GPU-batched sparse operations, which we leave to future engineering work. Overall, this confirms that the method's per-client and $O(K)$ server operations impose no barrier to industrial-scale deployment, while also surfacing genuine scale-dependent failure modes (naïve DP, adversarial instability) that a purely small-scale evaluation would have missed.

6 Trustworthiness Analysis and Regulatory Compliance

Composite trust score: read this caveat first.

Because a single headline number invites exactly the over-reading we want to avoid, we state its limits *before* presenting it rather than after. The composite score below is a *communication tool*: a compact way to say "good on every axis at once" that makes the axes and the normalisation explicit. It is not a certification of trustworthiness, not a substitute for the per-axis evidence, and not decisive evidence of superiority. Its ranking depends on choices we make—the reference constants, the unit weights, and the power p—and a reader who chooses different constants may obtain a different ordering. Auditors, deployers and reviewers should therefore treat the per-axis results (Tables 4–11, 9) as the primary empirical findings, and the composite as a summary of them. With that framing fixed, the construction is as follows.

Trustworthiness is multi-dimensional: a model that is accurate but unfair, or fair but privacy-leaking or attack-fragile, is not trustworthy. We summarise a method by a single interpretable score in $[0, 1]$ that aggregates *the available* normalised sub-scores—utility (AUC), fairness ($1 - \frac{\text{DPD+EOD}}{2}/F_{\text{ref}}$), privacy ($1 - \epsilon/\epsilon_{\max}$, and 0 **when no DP is applied**, so a non-private method is not rewarded by omission), calibration ($1 - \text{ECE}/\text{ECE}_{\text{ref}}$), and robustness (attack-retained AUC)—via a weighted power mean. The reference constants (F_{ref}=0.20, $\epsilon_{\max}$=16, ECE_{ref}=0.20) and unit weights are stated explicitly and are configurable, so that any reader can recompute the score under their own choices—which, per the caveat above, is exactly what we would encourage a sceptical auditor to do. Table 12 reports the arithmetic-mean (p=1) composite over the three axes computable from the main runs (utility, fairness, privacy); Table 11 adds the calibration axis on Bail; the robustness axis is reported separately (Table 9) rather than folded in, because it needs a matched clean/attacked pair. On the three-axis composite, TrustFedGNN attains the highest score, but the reason is specific and worth naming: it is the only entry that carries a genuine privacy guarantee (ϵ=8, privacy sub-score 0.5) *while* keeping utility and fairness high—every non-private baseline scores 0 on the privacy axis by construction. Read the composite as "good on every axis at once," not as a claim of being best on any single axis.

Explainability audit.

Because FSER retains its per-edge attention, its behaviour is inspectable. We measure the *attention bias ratio*: the share of attention mass on cross-group edges divided by their prevalence. A ratio below one—observed for TrustFedGNN—means the model attends to cross-group edges *less* than chance, which is direct evidence that FSER does what it is designed to do to the attention distribution. We are careful about the scope of this claim: it is a diagnostic of the *mechanism*, not a validated, faithful attribution of individual predictions, and we do not evaluate it against edge-masking or counterfactual-removal explainer benchmarks (a limitation, and a natural next step). Gradient attribution of the demographic-parity gap similarly localises which input features move the disparity, offering auditors a starting point for inspection rather than a certified explanation.

Table 12: Composite trust score on Bail: utility, fairness, and privacy combined via the arithmetic-mean (p=1) power mean, from the same canonical runs as Tables 4–6. Non-private methods score 0 on privacy by construction. The calibration-augmented composite is in Table 11. Ranking is sensitive to the reference constants (stated in text); best in bold.

Method	AUC	DPD	EOD	ϵ	Trust
FedAvg-GAT	0.972	0.070	0.024	–	0.579
FairGNN	0.906	0.047	0.017	–	0.582
FairSIN	0.956	0.065	0.025	–	0.577
FairFed	0.961	0.054	0.018	–	0.593
F^2GNN	0.971	0.064	0.021	–	0.586
DP-FedAvg	0.587	0.001	0.012	8	0.685
TrustFedGNN (no DP)	0.990	0.070	0.015	–	0.593
TrustFedGNN	0.988	0.066	0.009	8	**0.767**

Human-in-the-loop control.

Two budgets are exposed to a human operator: the fairness target τ (the operator-chosen cap on the reported client-disparity statistic that BFWA steers toward—see the scope caveat in Section 4.3) and the privacy budget ϵ (enforced by FTGD's accountant for the released statistic). The Pareto frontier (Figure 4) turns budget selection into an informed choice rather than a hidden default, supporting meaningful human oversight.

Regulatory alignment.

Table 13 offers a *design-level mapping* from TrustFedGNN's components to relevant EU AI Act articles (10, 13, 14, 15) and NIST AI Risk Management Framework functions (Govern/Map/Measure/Manage). We are deliberate that this is a design checklist showing which obligations each component is *intended to help address*—not a claim of legal compliance, which depends on system classification, provider/deployer role, jurisdiction, intended purpose, and a formal conformity assessment we do not perform. Concretely: raw graphs stay local and s is kept out of the input feature matrix (though, as noted in Section 4.1, FSER does use s inside its attention—a nuance an auditor must weigh); the FSER attention audit and model card support transparency; the τ, ϵ budgets support human oversight; and robust aggregation plus the DP guarantee speak to accuracy/robustness. The framework auto-generates a Mitchell-style model card [50], making these artifacts reproducible.

What requiring the protected attribute at inference actually costs a deployer.

This deserves more than the passing mention it received above, because it is the single design decision in TrustFedGNN with the largest practical and legal footprint. FSER's risk score $\phi_{ij} = \mathbb{I}(s_i \neq s_j) \cdot \max(0, \cos(\mathbf{h}_i, \mathbf{h}_j))$ (Eq. 4) evaluates the group indicator inside the forward pass. The model therefore needs s for *both endpoints of every edge it attends over*, not only during training but at every inference—and, because a decision about one customer aggregates over that customer's neighbourhood, an institution scoring a single applicant needs the protected attribute of the counterparties in that applicant's transaction neighbourhood too. Four consequences follow for a deploying institution. *First, lawful basis.* Gender, race and ethnicity are special-category data under GDPR Art. 9; processing them in an automated decision requires an Art. 9(2) condition, and in several EU member states the substantial-public- interest route (Art. 9(2)(g)) additionally needs a national-law basis. The EU AI Act anticipates exactly this tension: Art. 10(5) permits processing special-category data *for the purpose of bias detection and correction* in high-risk systems, subject to safeguards including technical limits on re-use, pseudonymisation, and deletion once bias correction is complete. FSER is squarely a bias-correction mechanism and so is the kind of processing that provision contemplates—but an institution must document that reliance rather than assume it, and Art. 10(5) is a permission to process, not a waiver of Art. 9 or of a DPIA. *Second, data-access agreements.* In the cross-silo setting the counterparties in a neighbourhood may be another institution's customers, so inference-time access to s can require inter-institutional agreements that plain FedAvg does not—a governance cost that partly offsets the "no raw data leaves the silo" benefit of federation. *Third, collection risk.* Where s is not already held, obtaining it means either new collection (consent, minimisation, retention limits) or *inference* of a protected attribute from proxies, which is itself a discriminatory-profiling risk and which we do not recommend. *Fourth, purpose limitation.* s must reach the attention mechanism without reaching the classifier as a feature: our implementation keeps it out of $\mathbf{X}$, but that separation is an engineering invariant a deployer must verify and an auditor must test, not a property the architecture enforces by itself. We consider these costs real but proportionate for the consortium setting we target, where a bias-audit lawful basis is typically already established for regulatory fairness reporting. For a deployer who cannot meet them, the honest answer is that FSER is the wrong component: the ablation (Table 7) shows FTGD and BFWA operating without it, at a fairness cost we quantify there. Removing the inference-time dependence entirely—by distilling FSER's behaviour into an s-free student model, or by replacing the hard indicator with a function of the DP-noised statistics—is the direction we consider most valuable for practical deployment, and we list it in Section 7.

7 Discussion, Limitations, and Broader Impact

Why FTGD works.

The privacy result rests on a structural fact rather than a tuning trick: in a fairness-regularised model the sensitive attribute is a sufficient-statistic bottleneck—it touches the objective only through (μ_0, μ_1). Privatising a two-dimensional statistic instead of a $|\theta|$-dimensional gradient avoids the $\sqrt{|\theta|}$ noise scaling that makes DP-SGD impractical on compact GNNs. The same principle applies to any group-fairness regulariser

Table 13: Mapping of TrustFedGNN components to EU AI Act articles and NIST AI RMF functions.

Requirement	Addressed by
EU AI Act	
Art. 10 - Data & data governance	Non-IID federated silos keep raw data local; sensitive attribute excluded from features; documented dataset provenance and base-rate gaps.
Art. 13 - Transparency & information	Per-edge FSER attention and feature-level disparity attribution expose why a decision is made; model card documents intended use and metrics.
Art. 14 - Human oversight	Explicit, human-set fairness budget tau and privacy budget epsilon; Pareto trade-off curves let an operator choose the operating point.
Art. 15 - Accuracy, robustness & cybersecurity	Byzantine-robust aggregation (robust_bfwa) with reported accuracy retention under poisoning/fairness attacks; RDP privacy guarantee.
NIST AI RMF	
GOVERN	Fairness/privacy budgets and audit logging make governance choices explicit and recorded.
MAP	Threat model (Byzantine + fairness poisoning) and subgroup-disparity context are documented.
MEASURE	DPD/EOD/EO, RDP epsilon, attack-retention, ECE and a composite trust score are all measured.
MANAGE	Robust aggregation + fairness constraint actively mitigate measured risks each round.

expressible through low-dimensional group statistics, so FTGD is reusable beyond GNNs.

Relation to the closest 2025–2026 methods.

It is worth stating precisely how TrustFedGNN differs from the most recent neighbours, because superficially several sound similar. F^{2}GNN and FairGFL are federated fair-graph methods, but the former offers no privacy and the latter targets *cross-client* performance fairness (equalising accuracy across silos with imbalanced overlaps) rather than demographic group fairness—a different objective that our BFWA does not attempt to displace. FaVGNN is the closest in ambition (federated, graph, group-fair, private), but it is formulated for *vertical* FL, where clients hold disjoint feature columns for shared entities; our horizontal cross-silo setting, where clients hold disjoint entities with the full feature set, is both more common in inter-bank fraud detection and structurally different, so we can only port FaVGNN's transferable client-side ideas rather than run it natively. FDP-Fair and PUFFLE combine privacy with fairness, but privatise the full gradient (or train with DP-SGD) and then repair fairness by post-processing or regularisation; FTGD instead *avoids* the full-gradient noise entirely by privatising the low-dimensional fairness statistic, which is why it retains utility where they must trade it away. Finally, and decisively, none of these methods addresses Byzantine robustness: a single malicious silo can corrupt any of them, whereas TrustFedGNN-Robust screens such updates before enforcing fairness. The contribution is thus not a marginal gain on one axis but the first coherent occupation of the intersection of all of them.

Limitations.

We are deliberately expansive here, because several of these bound the strength of our claims. (1) *Scope of the privacy guarantee.* FTGD's (ϵ, δ)-DP is over the *released fairness statistic* $(\tilde{\mu}_0, \tilde{\mu}_1)$, not over the whole transmitted update: the fairness-gradient masks and FSER's use of s leave sensitive-attribute dependence in the update that we do not privatise (Section 4.2). A fully update-level sensitive-attribute-DP mechanism is future work, and our comparison to DP-FedAvg is therefore not like-for-like. (2) *BFWA is a surrogate.* The constraint bounds a weighted average of *reported client* disparities, not the deployed model's DPD, which is a non-linear function of the aggregate; τ is an operator target, not a certified global bound, and can be infeasible—indeed several reported DPD values exceed the default τ=0.05 (Table 5), which we show rather than hide. (3) *DPD≠EOD.* We train on a soft demographic-parity term; this does not guarantee equal-opportunity improvement, and on Elliptic DPD improves while EOD *worsens* ($0.020 \rightarrow 0.075$)—a real tension we report. Demographic parity itself can be an inappropriate criterion when base rates genuinely differ across groups (recidivism, default), and should be chosen with domain and legal input rather than assumed. (4) *Statistical power, and where it runs out.* The main comparison is supported by paired Wilcoxon tests (Table 3) on the three datasets with at least five seeds—German (10), Bail (5), Pokec (5)—and those are the claims we regard as tested. Everything else is weaker and should be read accordingly: Credit and Elliptic use 2–3 seeds, so their columns carry means and standard deviations but no significance test; and the *robustness* (Table 9), large-scale (Table 10) and calibration (Table 11) studies are **single-seed**. This matters most for robustness, which is one of our contributions: we therefore restrict the robustness claims to order-of-magnitude effects (which aggregators collapse, and whether fairness is captured by the poisoning attack) and make no ranking claim among the aggregators that survive, whose differences are within plausible seed noise (Section 5.8). A multi-seed replication of the full aggregator–attack grid with error bars is the most valuable single addition to this evaluation and is planned future work. (4b) *Baseline coverage is uneven*: the full fifteen-baseline comparison is run only on German, Bail and Pokec; Credit uses eight baselines and Elliptic two (Section 5.1), so cross-method statements about those two datasets are correspondingly narrower. (5) *Binary s*; multi-valued/intersectional attributes are handled only by one-vs-rest, not exhaustively evaluated. (6) *Elliptic/ogbn-products subgroups are temporal/structural proxies*, not protected demographics; those results are subgroup-disparity analyses. (7) *Cross-client edges are dropped* (each silo is an induced subgraph), so this is federated simulation over a partitioned benchmark, not genuine multi-institution graph learning; inter-silo links under privacy are future work. (8) *Adapted baselines.* Several competitors are non-trivially adapted to our single-graph, star-topology setting (Appendix B); their numbers reflect our faithful-as-possible reimplementations, not necessarily the original papers, and should not be read as definitive rankings of those methods. (9) *Robustness assumes a known corruption count f* and a representative (non-adaptive) attack suite; misspecified f, adaptive attackers with knowledge of the screen, and the interaction between distance screening and honest non-IID minority clients are open issues. (10) *The composite trust score* is sensitive to its reference constants (F_{ref}, $\epsilon_{\max}$, $\mathrm{ECE}_{\mathrm{ref}}$), to its unit weights, and to the power p; a different, equally defensible choice of constants can reorder the methods. It is a communication tool and must not be read as a certification of trustworthiness or as evidence of superiority on any axis: the per-axis tables are the primary evidence, and we now say so before the score is introduced (Section 6) as well as here. It also covers only the axes computable from a given run—the three-axis composite of Table 12 omits calibration and robustness entirely—so a high score is a statement about the axes included, not about the system as a whole. (11) *Explainability* rests on attention/gradient diagnostics that we have not validated against faithfulness benchmarks. (12) *FSER requires the protected attribute at inference*, for both endpoints of every attended edge, which imposes lawful-basis, data-access and purpose-limitation obligations on a deploying institution that a sensitive-attribute-free model would not (Section 6). Removing this dependence—by distillation into an s-free student, or by replacing the hard group indicator with a function of the DP-noised statistics—is open work, and until it is done TrustFedGNN is only deployable where such processing is lawful and governed.

Broader impact.

Fraud, credit, and AML systems make consequential decisions about people. Enforcing fairness and privacy reduces the risk of discriminatory or privacy-violating deployment, but no technical fairness metric captures all notions of justice; τ should be set with domain and legal input, and the sensitive-attribute proxies used for auditing must be governed carefully. By releasing a reproducible, regulation-aligned framework we aim to lower the barrier to responsible deployment (SDG 9) rather than to certify any particular system as "fair".

Future work.

Several extensions follow naturally. The statistic-privatisation principle behind FTGD applies to any fairness criterion expressible through low-dimensional group statistics—equalised odds, sufficiency, calibration within groups—so a multi-criterion private-fairness framework is within reach. Handling *multi-valued and intersectional* sensitive attributes beyond the one-vs-rest reduction, and modelling *cross-silo edges* under privacy (where an entity's neighbours span institutions), are the two most consequential graph-specific directions. A third, driven by the deployment analysis of Section 6, is to remove FSER's inference-time dependence on the protected attribute—distilling the FSER-trained model into an s-free student, or driving the edge correction from the DP-noised statistics rather than a hard group indicator—which would materially widen the set of institutions that can lawfully deploy the framework. On robustness, replicating the aggregator–attack grid over multiple seeds is the immediate empirical priority; beyond it, adaptive attackers with full knowledge of the screening rule motivate a game-theoretic treatment, and combining our fairness constraint with certified robust aggregation is open. Replacing our MC-dropout calibration with a distribution-free conformal-prediction layer in the style of FedWQ-CP [43] is an attractive complementary direction, since it would give the composite trust score a finite-sample coverage guarantee rather than an asymptotic one. Finally, moving from a documented temporal proxy on Elliptic to genuine protected attributes on a consented financial dataset would let the fairness claims graduate from subgroup-disparity analysis to demographic guarantees.

Reproducibility.

All datasets are public and downloaded on demand; all randomness is seeded; privacy accounting, metrics, and aggregation rules are unit-tested in continuous integration; and every table and figure in this paper is regenerated from logged runs by a single script. The code, configurations, and logs are released so that each reported number is traceable to an exact configuration.

8 Conclusion

We presented TrustFedGNN, to our knowledge the first framework to integrate group fairness, differential privacy, and Byzantine robustness for federated graph neural networks in a single system, together with a trustworthiness instrumentation layer—predictive uncertainty, attention audits, a composite trust score, sustainability accounting, and an EU AI Act / NIST RMF design mapping. Its three components act at different points: FSER suppresses a targeted pattern of biased message passing, FTGD releases the low-dimensional fairness statistic under differential privacy at $O(1)$ noise cost (a statistic-level guarantee, not whole-update DP), and BFWA (with a Byzantine-robust variant) steers aggregation toward an operator-chosen disparity target while resisting poisoning, including a fairness-poisoning attack we introduce. Across five real datasets spanning credit, recidivism, social, and large-scale crypto domains, TrustFedGNN does not dominate every individual metric—specialised baselines win specific cells, and we report this openly—but it is the only method among fifteen baselines to combine competitive fairness and utility with statistic-level sensitive-attribute DP and Byzantine robustness at once, and every reported number is reproducible from released code and logs. We were candid about the limits: BFWA's constraint is a surrogate for global fairness rather than a certificate, the privacy guarantee covers the released statistic and not the full update, demographic parity does not entail equalised odds, FSER needs the protected attribute at inference and so carries real governance obligations, and several studies—robustness among them—remain single-seed. Future work will tighten these —update-level sensitive-attribute DP, a certified global-fairness bound, cross-silo edges under privacy, multi-valued attributes, and adaptive-attacker robustness—extending trustworthy federated graph learning toward real deployment.

Supplementary information. The full source code, configuration files, and experiment logs underlying every table and figure in this paper are released alongside the manuscript (see Data availability below).

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Declarations

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Conflict of interest. The authors declare that they have no competing interests.

Ethical approval. Not applicable. This study involved no human participants and no animal subjects. It uses only publicly released, de-identified benchmark datasets (German Credit, Credit Defaulter, Bail/Recidivism, Pokec, Elliptic, and ogbn-products), each obtained from its published distribution and used in accordance with its licence and intended research purpose. No ethics-committee approval was therefore required.

Consent to participate. Not applicable. No human participants were recruited or enrolled in this study, and no personal data were collected by the authors; all datasets analysed were previously published in de-identified form by their original curators.

Consent to publish. Not applicable. This article contains no data, images, or details relating to an identifiable individual person.

Data availability. All datasets analysed during the current study are publicly available. The German Credit, Credit Defaulter, and Bail (Recidivism) graphs and the Pokec-z social graph are available from the NIFTY and FairGNN benchmark releases (https://github.com/chirag126/nifty and https://github.com/EnyanDai/FairGNN); the Elliptic Bitcoin transaction graph is available from its public release on Kaggle (https://www.kaggle.com/datasets/ellipticco/elliptic-data-set); and ogbn-products is available from the Open Graph Benchmark (https://ogb.stanford.edu/docs/nodeprop/). No new data were generated in this study. The experiment logs from which every table and figure in this article is regenerated are included in the code repository cited below.

Materials availability. Not applicable.

Code availability. The complete source code, experiment configurations, and logged results supporting this article are publicly available at https://github.com/vinhqdang/FedFairGNN. Every table and figure can be regenerated from the released logs with a single script (`experiments/report.py`), as described in Appendix D.

Author contributions. Q.-V.D. and N.-S.-A.N. conceived the original FedFairGNN framework and the preliminary study. All authors contributed to extending the methodology, experiments, and analysis presented in this article, and all authors read and approved the final manuscript.

Appendix A Hyperparameters and training protocol

Table A1 lists every hyperparameter used, with the value and the range we explored. All methods share the backbone, optimiser, and federated budget; method-specific hyperparameters (adversary learning rate for FairGNN, q for q-FedAvg, β-step for FairFed, post-processing for FDP-Fair) use the values recommended by their originating papers. Seeds are fixed ($\{0, 1, 2\}$); differentially private runs are intrinsically stochastic, and we therefore report mean $\pm$ standard deviation.

Table A1: Hyperparameters. "Explored" gives the sweep range where applicable.

Hyperparameter	Default	Explored
GNN layers L	2	$\{2, 3\}$
Hidden dimension d_h	64	$\{32, 64, 128\}$
Attention heads (FSER/GAT)	4	$\{1, 4\}$
Dropout	0.3	$\{0.2, 0.3, 0.5\}$
Local optimiser	AdamW	—
Local learning rate η	0.01	$\{10^{-3}, 10^{-2}\}$
Weight decay	10^{-5}	—
Local epochs E	2	$\{1, 2, 3\}$
Communication rounds T	40–60	per dataset
Clients K	3–10	$\{3, 5, 10, 20\}$
Partition	Dirichlet(α=0.5)	$\alpha \in \{0.1, 0.5, 1.0\}$
Fairness weight λ	1.0	$\{0, 0.25, 0.5, 1, 2, 4\}$
Fairness budget τ	0.05	$\{0.01, 0.05, 0.1\}$
DP ϵ	8	$\{0.5, 1, 2, 4, 8, 16\}$
DP δ	10^{-5}	—
DP clip C	1.0	—
Frank–Wolfe iters T_{FW}	20	—
Dual step η_μ	0.1	—
MC-dropout samples	20	—

Appendix B Baseline reimplementation fidelity

Five 2025–2026 competitors (FairGFL [24], FedGraph-Fair [25], PUFFLE [19], FedFACT [20], and the PoPETs'25 FairFed variant [21]) were reimplemented from their published algorithm descriptions rather than from released code (none ship a public graph-federated implementation). Each paper's mechanism only partially transfers to this paper's single global-model, star-topology, Dirichlet-partitioned-single-graph setting; Table B2 states exactly what was kept and what was dropped or approximated for each, so that no reported number is over-claimed relative to the original method. FairGFL, FaVGNN, and FDP-Fair are likewise reimplemented (Section 2); full fidelity notes for all reimplemented baselines, including the exact equations reproduced from each source, are maintained alongside the code (`docs/BASELINES_AND_SOURCES.md`).

Table B2: What each 2025–2026 baseline's reimplementation keeps vs. drops.

Baseline	Kept	Dropped / approximated
FairGFL	weight $\propto 1/(1+O_i)$ aggregation rule	O_i (multi-graph node/edge overlap) proxied by sample-count deviation from the mean, since our setting has one Dirichlet-partitioned graph, not separate per-client graphs
FedGraph-Fair	minimax/DRO dual λ reweighting high-loss clients	personalised per-client models mixed via a learned peer-similarity graph (orthogonal decentralised-communication contribution)
PUFFLE	momentum controller auto-tuning the fairness weight toward a target disparity, + DP-SGD	the third privacy channel (noised group-count sharing for clients missing a demographic group) – unneeded here since every client observes both groups
FedFACT	closed-form global+local offset (exact reduction of the paper's optimum for binary demographic parity)	general multiclass cost-matrix calibration and the iterative dual-ascent solver
PoPETs'25	degree-2-polynomial FairFed weighting (the FHE-friendly surrogate itself)	threshold-CKKS secure aggregation and its noise-as-DP analysis (cryptographic infrastructure, no numeric effect once computed in the clear)

Appendix C Dataset construction and sensitive attributes

German, Credit, Bail follow the standard NIFTY preprocessing: edges are symmetrised, features are z-score standardised, and the sensitive attribute is excluded from the node feature matrix (so the model cannot read it directly). Sensitive attributes are gender (German), age (Credit), and race (Bail); targets are good-credit, no-default, and (non-)recidivism respectively. **Pokec** uses region as the sensitive attribute and working-field as the target. **Elliptic** is the public Bitcoin transaction graph; illicit transactions are the positive class, and the unlabelled majority of nodes is retained for message passing but excluded from loss and metrics. As Elliptic carries no protected demographic attribute, we construct a transparent operational proxy: early-versus late-period transactions, split at the median of the 49 anonymised time steps, so that we study whether an AML model disproportionately flags one market period—a genuine temporal-drift concern in deployed compliance systems. We stress this is a subgroup-disparity analysis, not a protected-demographic claim.

ogbn-products is used for the million-node scaling study (Section 5.10). It carries no protected demographic attribute, so we construct documented structural proxies rather than treat any column as a sensitive attribute: the binary target greedily coarsens the 47-way product category into two balanced super-classes (largest categories assigned to the positive class until $\approx 50\%$ of nodes are covered), giving a task a GNN can genuinely learn while remaining a deterministic function of the official label—not an arbitrary split; the sensitive attribute is a high- vs. low-degree split at the median node degree (a structural connectivity subgroup, analogous to core–periphery position in a co-purchase graph). Training uses neighbor sampling (fan-out $10, 5$; batch size 4096); the official OGB split is used for train/validation/test. The test split alone contains 2.2M nodes, so both per-round monitoring and the final reported metrics evaluate a fixed random sub-sample of 100k test nodes (seeded, so identical across rounds and comparable across methods) rather than the full test set, keeping evaluation cost from dominating training cost at this scale.

Clients receive induced subgraphs under a Dirichlet(α) label partition with a small per-client class floor that prevents degenerate single-class silos; each client inherits the global train/validation/test masks for its nodes, and the global model is evaluated on the pooled test nodes across all clients.

Appendix D Reproducibility

All datasets are public and downloaded on demand into a local cache; all randomness is seeded; the privacy accountant, metrics, and aggregation rules are covered by an offline unit-test suite run in continuous integration; and every table and figure in this paper is regenerated from logged runs by a single script (`experiments/report.py`). The full experiment matrix—main comparison, ablation, privacy sweep, fairness–utility Pareto frontier, Byzantine-robustness study, scalability, and partition-sensitivity—is driven by a resumable runner (`experiments/run_matrix.py`). Code, configuration, and logs are released at https://github.com/vinhqdang/FedFairGNN.

References

[1] Kipf, T.N., Welling, M.: Semi-supervised classification with graph convolutional networks. In: International Conference on Learning Representations (ICLR) (2017). https://openreview.net/forum?id=SJU4ayYgl

[2] Veličković, P., Cucurull, G., Casanova, A., Romero, A., Liò, P., Bengio, Y.: Graph attention networks. In: International Conference on Learning Representations (ICLR) (2018). https://openreview.net/forum?id=rJXMpikCZ

[3] Dou, Y., Liu, Z., Sun, L., Deng, Y., Peng, H., Yu, P.S.: Enhancing graph neural

network-based fraud detectors against camouflaged fraudsters. In: ACM International Conference on Information and Knowledge Management (CIKM), pp. 315–324 (2020). https://doi.org/10.1145/3340531.3411903

[4] McMahan, B., Moore, E., Ramage, D., Hampson, S., Arcas, B.: Communication-efficient learning of deep networks from decentralized data. In: Artificial Intelligence and Statistics (AISTATS), vol. 54, pp. 1273–1282 (2017). https://proceedings.mlr.press/v54/mcmahan17a.html

[5] Dai, E., Wang, S.: Say no to the discrimination: Learning fair graph neural networks with limited sensitive attribute information. In: ACM International Conference on Web Search and Data Mining (WSDM), pp. 680–688 (2021). https://doi.org/10.1145/3437963.3441752

[6] Yang, C., Liu, J., Yan, Y., Shi, C.: FairSIN: Achieving fairness in graph neural networks through sensitive information neutralization. In: AAAI Conference on Artificial Intelligence, vol. 38, pp. 9241–9249 (2024). https://doi.org/10.1609/aaai.v38i8.28776

[7] Lee, Y.-C., Shin, H., Kim, S.-W.: Disentangling, amplifying, and debiasing: Learning disentangled representations for fair graph neural networks. In: AAAI Conference on Artificial Intelligence, vol. 39, pp. 12013–12021 (2025). https://doi.org/10.1609/aaai.v39i11.33308

[8] Ezzeldin, Y.H., Yan, S., He, C., Ferrara, E., Avestimehr, A.S.: FairFed: Enabling group fairness in federated learning. In: AAAI Conference on Artificial Intelligence, vol. 37, pp. 7494–7502 (2023). https://doi.org/10.1609/aaai.v37i6.25911

[9] Li, T., Sanjabi, M., Beirami, A., Smith, V.: Fair resource allocation in federated learning. In: International Conference on Learning Representations (ICLR) (2020). https://openreview.net/forum?id=ByexElSYDr

[10] Blanchard, P., El Mhamdi, E.M., Guerraoui, R., Stainer, J.: Machine learning with adversaries: Byzantine tolerant gradient descent. In: Advances in Neural Information Processing Systems (NeurIPS), pp. 119–129 (2017). https://proceedings.neurips.cc/paper/2017/hash/f4b9ec30ad9f68f89b29639786cb62ef-Abstract.html

[11] Yin, D., Chen, Y., Ramchandran, K., Bartlett, P.L.: Byzantine-robust distributed learning: Towards optimal statistical rates. In: International Conference on Machine Learning (ICML), vol. 80 (2018). https://proceedings.mlr.press/v80/yin18a.html

[12] Kasyap, H., Fang, M., Liu, Z., Maple, C., Tripathy, S.: Fairness-Constrained Optimization Attack in Federated Learning. arXiv preprint arXiv:2510.12143 (2025). https://arxiv.org/abs/2510.12143

[13] Dang, Q.-V., Nguyen, N.-S.-A.: FedFairGNN: A privacy-preserving and fairness-aware federated graph neural network for fraud detection. In: Proceedings of IndabaX Nigeria 2026: Building Scalable AI That Works: From Research to Deployment in Resource-Constrained Environments. Proceedings of Machine Learning Research, vol. 319, pp. 74–86 (2026). https://proceedings.mlr.press/v319/dang26a.html

[14] Agarwal, C., Lakkaraju, H., Zitnik, M.: Towards a unified framework for fair and stable graph representation learning. In: Uncertainty in Artificial Intelligence (UAI), vol. 161, pp. 2114–2124 (2021). https://proceedings.mlr.press/v161/agarwal21b.html

[15] Dong, Y., Liu, N., Jalaian, B., Li, J.: EDITS: Modeling and mitigating data bias for graph neural networks. In: The Web Conference (WWW), pp. 1259–1269 (2022). https://doi.org/10.1145/3485447.3512173

[16] Li, Z., Dong, Y., Liu, Q., Yu, J.X.: Rethinking fair graph neural networks from re-balancing. In: ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), pp. 1736–1745 (2024). https://doi.org/10.1145/3637528.3671826

[17] Zhu, Y., Li, J., Bian, Y., Zheng, Z., Chen, L.: One fits all: Learning fair graph neural networks for various sensitive attributes. In: ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD) (2024). https://doi.org/10.1145/3637528.3672029

[18] Park, S., Lim, S.: FnRGNN: Distribution-aware fairness in graph neural network. In: ACM International Conference on Information and Knowledge Management (CIKM) (2025). https://doi.org/10.1145/3746252.3760796

[19] Corbucci, L., Heikkilä, M.A., Solans Noguero, D., Monreale, A., Kourtellis, N.: PUFFLE: Balancing privacy, utility, and fairness in federated learning. In: European Conference on Artificial Intelligence (ECAI). Frontiers in Artificial Intelligence and Applications, vol. 392, pp. 1639–1647 (2024). https://doi.org/10.3233/FAIA240671

[20] Zhang, L., Han, Z., Feng, X., Zhang, J., Li, Y., Chen, C.: FedFACT: A provable framework for controllable group-fairness calibration in federated learning. In: Advances in Neural Information Processing Systems (NeurIPS) (2025). https://arxiv.org/abs/2506.03777

[21] Bendoukha, A.-A., Demirag, D., Kaaniche, N., Boudguiga, A., Sirdey, R., Gambs, S.: Towards privacy-preserving and fairness-aware federated learning framework. Proceedings on Privacy Enhancing Technologies (PoPETs) **2025**(1), 845–865 (2025) https://doi.org/10.56553/popets-2025-0044

[22] Yu, T., Zhao, K., Zhang, C., Gao, A., Quan, Y., Liu, Z., Fang, M.: When the Server Steps In: Calibrated Updates for Fair Federated Learning. arXiv preprint

arXiv:2601.05352 (2026). https://arxiv.org/abs/2601.05352

[23] Cui, N., Wang, X., Wang, W.H., Chen, V., Ning, Y.: Equipping federated graph neural networks with structure-aware group fairness. In: IEEE International Conference on Data Mining (ICDM) (2023). https://doi.org/10.1109/ICDM58522.2023.00111

[24] Zhou, Z., Yang, S., Zhao, F., Ren, X.: FairGFL: Privacy-preserving fairness-aware federated learning with overlapping subgraphs. IEEE Transactions on Parallel and Distributed Systems **37**(3) (2026) https://doi.org/10.1109/TPDS.2025.3649863

[25] Khan, K.: FedGraph-Fair: Federated learning with personalization and fairness via dynamic graphs and distributionally robust optimization. Information Sciences **728**, 122710 (2026) https://doi.org/10.1016/j.ins.2025.122710

[26] Wang, Z., Yan, X., Jin, Y.: Fairness-oriented vertical federated GNNs with incomplete sensitive attributes. Information Fusion **127**, 103871 (2026) https://doi.org/10.1016/j.inffus.2025.103871

[27] Xue, G., Yu, Y.: Federated Fairness-Aware Classification under Differential Privacy. arXiv preprint arXiv:2603.24392 (2026). https://arxiv.org/abs/2603.24392

[28] McMahan, H.B., Ramage, D., Talwar, K., Zhang, L.: Learning differentially private recurrent language models. In: International Conference on Learning Representations (ICLR) (2018). https://openreview.net/forum?id=BJ0hF1Z0b

[29] Mironov, I.: Rényi differential privacy. In: IEEE Computer Security Foundations Symposium (CSF), pp. 263–275 (2017). https://doi.org/10.1109/CSF.2017.11

[30] Abadi, M., Chu, A., Goodfellow, I., McMahan, H.B., Mironov, I., Talwar, K., Zhang, L.: Deep learning with differential privacy. In: ACM SIGSAC Conference on Computer and Communications Security (CCS), pp. 308–318 (2016). https://doi.org/10.1145/2976749.2978318

[31] Ling, X., Fu, J., Wang, K., Li, H., Cheng, T., Chen, Z.: FedFDP: Fairness-Aware Federated Learning with Differential Privacy. arXiv preprint arXiv:2402.16028 (2024). https://arxiv.org/abs/2402.16028

[32] Guo, Z., Wu, W., Chen, C., Zhang, L., Kotevska, O., Madduri, R.K.: Provably Communication-Efficient and Privacy-Preserving Federated Graph Neural Networks. arXiv preprint arXiv:2605.26243 (2026). https://arxiv.org/abs/2605.26243

[33] Varun, M., Dhakar, A.K., Hong, Y., Sural, S.: Adversarial Attacks on Locally Private Graph Neural Networks. arXiv preprint arXiv:2603.20746 (2026). https://arxiv.org/abs/2603.20746

[34] El Mhamdi, E.M., Guerraoui, R., Rouault, S.: The hidden vulnerability of distributed learning in byzantium. In: International Conference on Machine Learning (ICML), vol. 80, pp. 3521–3530 (2018). https://proceedings.mlr.press/v80/mhamdi18a.html

[35] Cao, X., Fang, M., Liu, J., Gong, N.Z.: FLTrust: Byzantine-robust federated learning via trust bootstrapping. In: Network and Distributed System Security Symposium (NDSS) (2021). https://doi.org/10.14722/ndss.2021.24434

[36] Bagdasaryan, E., Veit, A., Hua, Y., Estrin, D., Shmatikov, V.: How to backdoor federated learning. In: Artificial Intelligence and Statistics (AISTATS), vol. 108, pp. 2938–2948 (2020). https://proceedings.mlr.press/v108/bagdasaryan20a.html

[37] Baruch, G., Baruch, M., Goldberg, Y.: A little is enough: Circumventing defenses for distributed learning. In: Advances in Neural Information Processing Systems (NeurIPS), pp. 8632–8642 (2019). https://proceedings.neurips.cc/paper/2019/hash/ec1c59141046cd1866bbbcdfb6ae31d4-Abstract.html

[38] Xie, C., Koyejo, O., Gupta, I.: Fall of empires: Breaking byzantine-tolerant SGD by inner product manipulation. In: Uncertainty in Artificial Intelligence (UAI), vol. 115, pp. 261–270 (2019). https://proceedings.mlr.press/v115/xie20a.html

[39] Meerza, S.I.A., Liu, J.: EAB-FL: Exacerbating algorithmic bias through model poisoning attacks in federated learning. In: International Joint Conference on Artificial Intelligence (IJCAI), pp. 458–466 (2024). https://doi.org/10.24963/ijcai.2024/51

[40] Nath, A., Kasyap, H.: CQSA: Byzantine-robust Clustered Quantum Secure Aggregation in Federated Learning. arXiv preprint arXiv:2602.22269 (2026). https://arxiv.org/abs/2602.22269

[41] Rahmati, M., Rahmati, N.: Byzantine-Robust Federated Learning Framework with Post-Quantum Secure Aggregation for Real-Time Threat Intelligence Sharing in Critical IoT Infrastructure. arXiv preprint arXiv:2601.01053 (2026). https://arxiv.org/abs/2601.01053

[42] Gal, Y., Ghahramani, Z.: Dropout as a bayesian approximation: Representing model uncertainty in deep learning. In: International Conference on Machine Learning (ICML), vol. 48, pp. 1050–1059 (2016). https://doi.org/10.5555/3045390.3045502

[43] Nguyen, Q.-H., Wang, J., Ku, W.-S.: Conformalized Neural Networks for Federated Uncertainty Quantification under Dual Heterogeneity. arXiv preprint arXiv:2602.23296 (2026). https://arxiv.org/abs/2602.23296

[44] Ali, M.H., Abdullah, M.A., Hussain, S.M., Uddin, M.M., Alam, S.: Privacy-Preserving Federated Learning with Verifiable Fairness Guarantees. arXiv preprint arXiv:2601.12447 (2026). https://arxiv.org/abs/2601.12447

[45] Hsu, T.-M.H., Qi, H., Brown, M.: Measuring the Effects of Non-Identical Data Distribution for Federated Visual Classification. arXiv preprint arXiv:1909.06335 (2019). https://arxiv.org/abs/1909.06335

[46] Dwork, C., Roth, A.: The algorithmic foundations of differential privacy. Foundations and Trends in Theoretical Computer Science **9**(3–4), 211–487 (2014) https://doi.org/10.1561/0400000042

[47] Jaggi, M.: Revisiting frank–wolfe: Projection-free sparse convex optimization. In: International Conference on Machine Learning (ICML), vol. 28, pp. 427–435 (2013). https://doi.org/10.5555/3042817.3042867

[48] Hu, W., Fey, M., Zitnik, M., Dong, Y., Ren, H., Liu, B., Catasta, M., Leskovec, J.: Open graph benchmark: Datasets for machine learning on graphs. In: Advances in Neural Information Processing Systems (NeurIPS) (2020). https://doi.org/10.48550/arXiv.2005.00687

[49] Hamilton, W.L., Ying, R., Leskovec, J.: Inductive representation learning on large graphs. In: Advances in Neural Information Processing Systems (NeurIPS) (2017). https://doi.org/10.48550/arXiv.1706.02216

[50] Mitchell, M., Wu, S., Zaldivar, A., Barnes, P., Vasserman, L., Hutchinson, B., Spitzer, E., Raji, I.D., Gebru, T.: Model cards for model reporting. In: ACM Conference on Fairness, Accountability, and Transparency (FAT*) (2019). https://doi.org/10.1145/3287560.3287596